



Sequential SI-ATRP in μL -scale for surface nanoengineering: A new concept for designing polyelectrolyte nanolayers formed by complex architecture polymers

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ABSTRACT

The article presents a step-by-step protocol, designed on the basis of principles of green chemistry, for the synthesis of polymer brushes with complex structure via surface-initiated atom transfer radical polymerization (SI-ATRP) approach. The selection of anionic (poly(acrylic acid), PAA) and cationic (poly(2-(dimethylamino)ethyl methacrylate, PDMAEMA) macromolecular chains enabled the formation of ampholytic polymer coating on silicon (Si) substrates. Scanning electron microscopy (SEM), atomic force microscopy (AFM) and X-ray photoelectron spectroscopy (XPS) analyses confirmed changes in the structure of the synthesized nanolayer at each step of the surface modification. Finally, the synthesis protocol was rescaled presenting an extremely economically and environmentally beneficial procedure for the synthesis of Si wafers grafted with brushes composed of poly(2-hydroxyethyl methacrylate) (PHEMA) core and PDMAEMA and PAA as block polymer branches (Si-g-PHEMA-g-(PDMAEMA-*b*-PAA)) with application of microliter volumes of reactants.

1. Introduction

One of the noteworthy ways of designing well-defined functional polymer coatings is the application of surface-initiated atom transfer radical polymerization (SI-ATRP) [1,2] as well as surface-initiated photoinduced electron transfer-reversible addition-fragmentation chain transfer polymerization (SI-PET-RAFT) [3]. The successively monomer-by-monomer built up chain segments result in polymers with narrow molecular weight distribution (MWD), well-defined molecular weights (MW) and predetermined composition [4]. Moreover, recently to overcome limitations in the modified area as well as reduce the mixture volume used, microliter-scale diffusion-controlled processes (μL -scale SI-ATRP) has been evolved [5,6]. Combining microliter volumes of reactants with low ppm techniques created an economical and environmentally friendly tool for effective material functionalization and bioconjugates synthesis [7,8]. What is remarkable, densely grafted polymer brushes with desired properties can be obtained using even a few μL of reaction mixture per 1 cm^2 of modified surface. A number of

advantages of the ATRP, led by its controlled nature, enable the direct synthesis of macromolecules with complex topology and molecular architecture, among others, bottlebrushes otherwise known as molecular brushes [9-11].

Considering the polymerization mechanism, the ATRP is based on the equilibrium between the propagating radicals, formed by homolytic cleavage of the C-X (where X – halogen atom) bond by a transition metal complex at a lower oxidation state ($\text{Cu}^{\text{I}}/\text{L}$), and the dormant alkyl halides, and the transfer of the X atom from the oxidized form of the complex ($\text{X-Cu}^{\text{II}}/\text{L}$) to the radical [12-14]. However, “classical” ATRP requires high concentrations of a transition metal, typically copper. Therefore, by introducing an additional redox cycle, copper concentration can be reduced to tens of ppm by continuous regeneration of the catalytic complex [15,16]. The widely used, internal reducing agents are zerovalent metals (Fe^0 [17-19], Cu^0 [20-22]) acting as a supplemental activator and a reducing agent (SARA ATRP) or substances like ascorbic acid (AsAc) [23-25] or glucose [26] in activators regenerated by electron transfer ATRP (ARGET ATRP) technique. In turn, external control of

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deactivator reduction might be achieved by applying an electric current in electrochemically mediated ATRP (eATRP) [27–31] or light in photomediated ATRP (photo-ATRP) [32–36].

Importantly, SI-ATRP makes it possible to graft multifunctional and stimuli-responsive polymer brushes such as polyelectrolytes and obtain anionic, cationic and zwitterionic surfaces in result [37,38]. Similarly to inorganic compounds, one can distinguish weak and strong polyelectrolytes, depending on the degree on their ionization in solutions [39]. Strong polyelectrolytes are fully dissociated in the majority of pH, whereas weak also occur in unionized form, depending on the environment nature [40]. Therefore, weak polyelectrolytes such as PAA or PDMAEMA might be characterized by the ability to response to changes in pH value and ionic strength by switching their conformation due to alterations in their H-bonding ability or electrostatic or hydrophobic interactions. Depending on the value of pKa, an increase in the ionization degree of the polyelectrolyte results in brush swelling, while a low ionization degree causes the polymer chains to collapse [41–43].

In recent years, relatively few articles on complex architecture polyelectrolyte brushes grafted from surfaces can be found [44]. One new paper reports the synthesis of cationic bottlebrush-like polymers on silicon wafers surface via SI-ATRP and ring-opening metathesis polymerization for potential use in solid-state batteries [45]. Therefore, in this paper, complex ampholytic polymer brushes fabricated via step-by-step SI-ATRP with a PHEMA backbone and PDMAEMA and PAA block copolymers side chains grafted from silicon wafers were presented and carefully characterized for the first time. Following the principles of green chemistry, a low ppm system with continuous regeneration of catalyst was applied (ARGET ATRP, SARA ATRP and dually controlled photo-ATRP). Finally, the consumption of reagents was reduced by using μL -scale SI-ATRP in an environmentally safe ethanol-based reaction medium.

2. Experimental

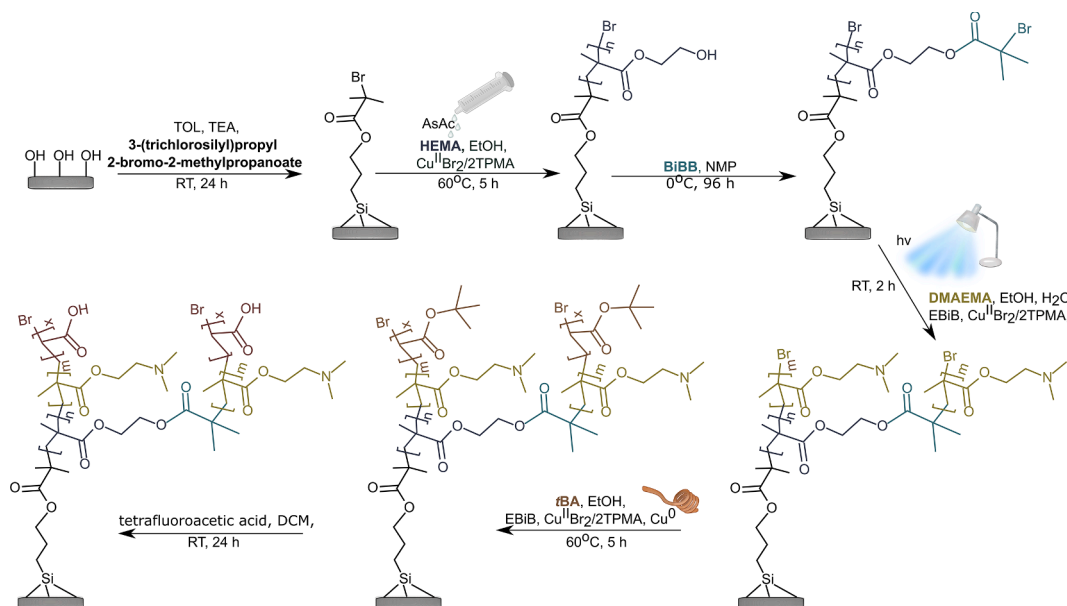
An experimental section including material characteristics, analyses, SI-ATRP procedures, sample cleaning protocols, real modified silicon wafers photos, water contact angle (WCA) results, free surface energy (FSE) data, ^1H NMR and ToF-SIMS spectra of polymers, EDS spectra and AFM images (AFM) were included in the [Supplementary information](#).

3. Results and discussion

Although the application of ATRP techniques to the synthesis of polyelectrolytes is a well-known solution, limited attention is paid to the possibility of developing environmentally friendly procedures by, among other things, reducing the concentration of the catalytic copper complex or promoting “greener” solvent than DMF or THF [46]. Therefore, these aspects were the main focus of the development of the concept for the synthesis of previously undescribed polymer brushes with complex architectures and a polyelectrolyte nature.

3.1. Synthesis of ampholytic polymer nanolayer via low ppm SI-ATRP methods

The synthesis of ampholytic polymer coatings was carried out using step-by-step SI-ATRP techniques with additional redox cycle (low ppm ATRP) allowing minimization of the concentration of copper(II) bromide/tris(2-pyridylmethyl)amine ($\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$) to the 281–221 ppm by wt. The selection of the low ppm ATRP techniques for the polymerizations of the monomers, presented in [Scheme 1](#), was developed on the basis of our own previous experimental works [47,48]. With a view to selecting the preferred technique that enables controlled polymerization of individual functional monomers, three various approaches as ARGET ATRP, photo-ATRP and SARA ATRP were used to synthesize the complex architecture polymer brushes. In general, there are many factors influencing the nature of ATRP, from the choice of physical quantities (such as e.g. synthesis temperature) [49] to the characteristics of each individual substance included in the reaction mixture [50,51]. Here, following the principles of green chemistry, where possible, techniques were chosen that limit the introduction of additional reactants (e.g., photo-ATRP for DMAEMA) or allow reuse of the reducing agent as in the case of the copper wire used in SARA ATRP of *t*BA. In the case of HEMA polymerization, special attention was focused on selecting a technique resulting in a polymer with low dispersity, with the goal of synthesizing a homogeneous brush layer that provided a core to be expanded in subsequent synthesis steps. Furthermore, due to the non-toxic nature of the ascorbic acid (classified as FDA-approved reagent) used to regenerate the Cu^{I} complex activators, ARGET ATRP can also be included in



Scheme 1. The assumed concept of “green” synthesis pathway of polyelectrolyte surface-grafted polymers with complex architecture. Abbreviations: DMAEMA – 2-(dimethylamino)ethyl methacrylate, HEMA – 2-hydroxyethyl methacrylate and *t*BA – *tert*-butyl acrylate as monomers, EtOH – ethanol, DCM – dichloromethane, NMP – *N*-methyl-2-pyrrolidone and TOL – toluene as solvents, AsAc – ascorbic acid and Cu^0 – copper wire as reducing agents, $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ – catalytic complex, where TPMA – tris(2-pyridylmethyl)amine, BiBB – α -bromoisobutyl bromide and EBIB – ethyl α -bromoisobutyrate as ATRP initiators, RT – room temperature, TEA – triethylamine.

the list of ATRP “green” approaches.

Having regard to the above, the construction of the macromolecular coatings with a complex structure included (a) attachment of the ATRP initiator to the surface of the silicon wafer, (b) polymerization of 2-hydroxyethyl methacrylate (HEMA) by the surface-initiated ARGET ATRP technique, (c) esterification of pendant hydroxyl groups in PHEMA block using α -bromoisobutyl bromide (BiBB), (d) grafting of PDMAEMA from the brominated PHEMA backbone by the double-controlled (internally and externally) blue light-mediated photo-ATRP, (e) chain-extension of the surface-grafted brush polymers with poly(*tert*-butyl acrylate) PtBA generated via SARA ATRP technique, and (f) hydrolysis of *tert*-butyl groups to carboxyl groups to obtain PAA block. Eventually, the controlled nature of ATRP made it feasible to easily prepare a molecule with a complex architecture and with preservation of the chain-end functionality (Scheme 1).

A summary of the ATRPs used to generate Si-g-HEMA-g-(PDMAEMA-*b*-PAA) were presented in Table 1. Due to the addition of a sacrificial initiator to the reaction mixture, data summarized in Table 1 refer to the homopolymers formed in solution. At the same time, according to Scheme 1, the initiation of polymerization from the surface of the silicon wafer resulted in complex architecture polymer brushes.

In the first stage, brominated silicon wafers with attached ATRP initiator were prepared as in previously published work [8]. The successful modification of Si with Br was confirmed by ToF-SIMS analysis, showing two peaks for *m/z* ratio equal to 79 and 81, which correspond to two stable Br isotopes (Fig. S1). Subsequently, optimization of PHEMA synthesis was carried out (see section: S2.2 Optimization of PHEMA synthesis conditions in Supplementary data Table S1, Fig. S2) including the reaction with the presence of ethyl α -bromoisobutyrate and without the addition of the free initiator to the solution (Supplementary information, Table S2). At first, based on previous experience, it was visually assessed that in the case of HEMA polymerizations, the optimal results (thick and homogeneous layer) were obtained after 5 h of ARGET ATRP with 250 ppm of Cu^{II}Br₂/TPMA in the absence of the sacrificial initiator in reaction mixture (as observed in the photo in Table S2, entry 5). Therefore, for the final step-by-step procedure, the SI-ATRP was conducted without any additional initiators in the solution, while ATRP initiator was exclusively present on the surface of silicon wafer (Table 1, entry 1). The applied protocol resulted in the formation of a homogeneous polymer layer with dry thickness (*h*_{dry}) around 84 nm (Fig. 1a). In the next step, following the concept of ATRP macroinitiator synthesis [52,53], the number of initiation sites was multiplied by esterification of hydroxyl moieties in the PHEMA chain with α -bromoisobutyl bromide. Importantly, an increase of the thickness of the polymer layer from 84 nm to 178 nm suggest that nearly quantitative modification of the hydroxyl groups was achieved assuming that the brush thickness increase is proportional to the change of total mass of the layer. In case of quantitative bromination, the percentage change of the total mass of the layer could be calculated by comparing the mass of HEMA mer with attached BiBB group (279 u) with unmodified mer (130 u). As a result, the percentage mass increase, in case of quantitative substitution, should be around 215% ($\frac{279-130}{130} \times 100\% = 214.6\%$), while the thickness should reach

180 nm. In our case the thickness increased by around 212% up to 178 nm, hence one may expect practically quantitative modification. In sequence, a dually-controlled photo-ATRP of DMAEMA was carried out (Table 1, entry 2) [47], and then using the advantages of ATRP techniques, the retained functionality of the final chain was confirmed by the addition of a PtBA segment (Table 1, entry 3) via SARA ATRP (reaction conditions selected after optimization included in Table S3), which might act as a polyanion after hydrolysis to PAA [48]. Additionally, the homopolymer brushes were synthesized using the same procedures to determine the thicknesses of homopolymer layers by AFM. The thickness of the obtained homopolymer layers was equal 28 nm for silicon wafer grafted with PDMAEMA (Si-g-PDMAEMA) (Fig. S4a) and 27 nm for silicon wafer grafted with PtBA (Si-g-PtBA) (Fig. S4b). The appearance of the silicon wafers after each synthesis step (Scheme 1) indicated changes in the thickness of the polymer layer as well – the real photographs of plates after modifications were summarized in supporting information as Fig. S5. The direct correlation between the color changes observed in the photos and the increase in brush thickness after each surface treatment stage (except hydrolysis which resulted in a decrease of *h*_{dry}) was confirmed by AFM results (Fig. 1). It should be noted that after the addition of PtBA chains, a bit of roughness were observed on the surface of Si-g-PHEMA-g-(PDMAEMA-*b*-PtBA), which may suggest the occurrence of surface nanoscale phase separation (Fig. 1d). This type of phenomenon can be observed in the case of combining polymers that are (in)compatible (immiscible) [54]. Nevertheless, after hydrolysis, the phenomenon was limited due to conversion of PtBA segment to hydrophilic PAA (Fig. 1e).

Although the assumed synthesis pathway (Scheme 1) was inspired by typical procedures for the synthesis of bottlebrushes generated in solution, it is assumed that the steric hindrance at the wafer surface might limit the possibility of grafting side PDMAEMA and PtBA chains from the esterified PHEMA. Such suspicion was suggested by the lower-than-expected *h*_{dry} determined for Si-g-PHEMA-g-PDMAEMA, where the dry thickness of polymer layer increased by only 23 nm in comparison to Si-g-PHEMA-Br (Fig. 1b vs. Fig. 1c). Therefore, it seems that the actual complex architecture of the obtained polymers mainly consist of block copolymer grafted on the top of the main PHEMA backbone (Si-g-PHEMA-Br) as shown in Scheme 2. However, one may also expect the appearance of the bottlebrush structure near the chain ends of the PHEMA. In order to confirm that assumptions, the synthesized hybrid material was subjected to the detailed characterization as described in Section 3.2.

As repeatedly reported for water-based conditions, the high values of the ATRP equilibrium constants (*K*_{ATRP}) and the activation rate coefficients (*k*_{act}) as well as the limited stability of the activator and the deactivator, and consequently the occurrence of dissociation and disproportionation, led to extremely fast polymerization with limited control [55]. Therefore the use of a mixture of ethanol and water was proposed as a solution that allows to accelerate the photo-ATRP of DMAEMA while maintaining its controlled character [56]. The ¹H NMR data for the sacrificial initiator-assisted syntheses (Table 1, entry 2–3) confirmed the controlled nature of polymerization. In turn, data

Table 1

Summary of homopolymer chains generated in solution during sacrificial initiator-assisted low ppm SI-ATRP applied to generate polyelectrolyte coating.¹

Entry	Type of reaction	Monomer	Solvent	Conv ² [%]	<i>k</i> _p ^{app 2} [h ⁻¹]	<i>M</i> _{n,theo} ³	<i>M</i> _{n,app} ⁴	<i>D</i> ⁴
1	ARGET ATRP ⁵	HEMA	EtOH	–	–	–	–	–
2	photo-ATRP	DMAEMA	EtOH/H ₂ O	27	0.14	101,400	82,000	1.45
3	SARA ATRP	tBA	EtOH	44	0.15	56,100	39,400	1.21

¹ General reaction conditions: T = RT (except entry 3: T = 60 °C); V_{tot} = 18 mL (except entry 3: V_{tot} = 9) (in monomer/solvent = 1/1 by v/v) [I]₀ = 3.4 mM (except entry 1: no sacrificial initiator); [HEMA]₀ = 4.1 M; [DMAEMA]₀ = 3.0 M; [tBA]₀ = 3.4 M; for entry 2 DP_{target} = 2,400 and for entry 3 DP_{target} = 1,000. 2-Fold excess of TPMA was used to stabilize the complex. ² Monomer conversion (Conv), theoretical *M*_n and apparent propagation rate coefficient (*k*_p^{app}) were determined by ¹H NMR. *k*_p^{app} was determined as a slope of the line ln[M]₀/[M] = f(t) illustrated in the Fig. 2a and 2d; ³*M*_{n, th} = ([M]₀/[I]₀) × Conv × *M*_{monomer} + *M*_{initiator}; ⁴*M*_{n,app} and *D* was determined by DMF + 50 mM LiCl GPC with PS standards. ⁵ synthesis without sacrificial initiator, therefore neither conversion nor *M*_{n,th}, *M*_{n,app} and *D* were determined.

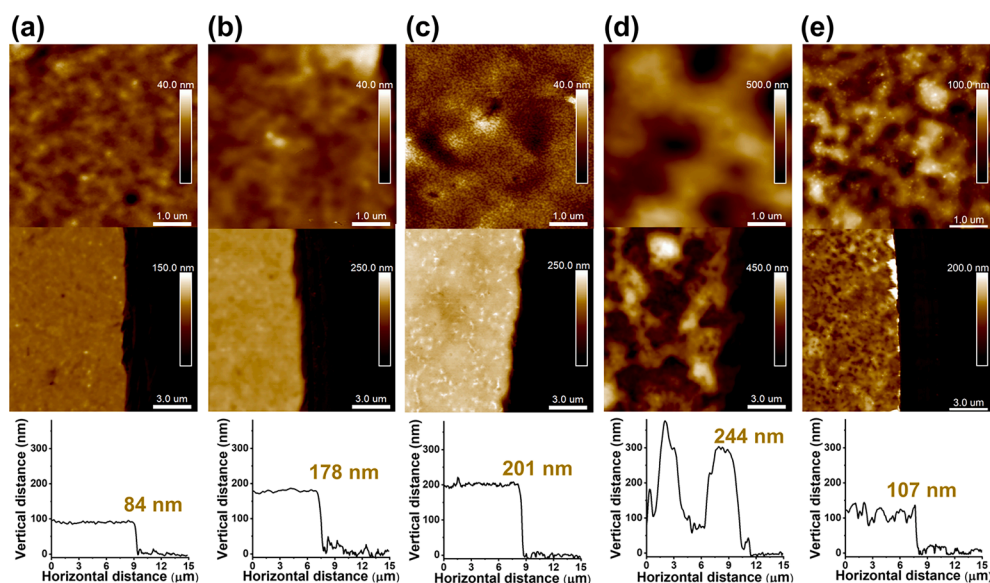
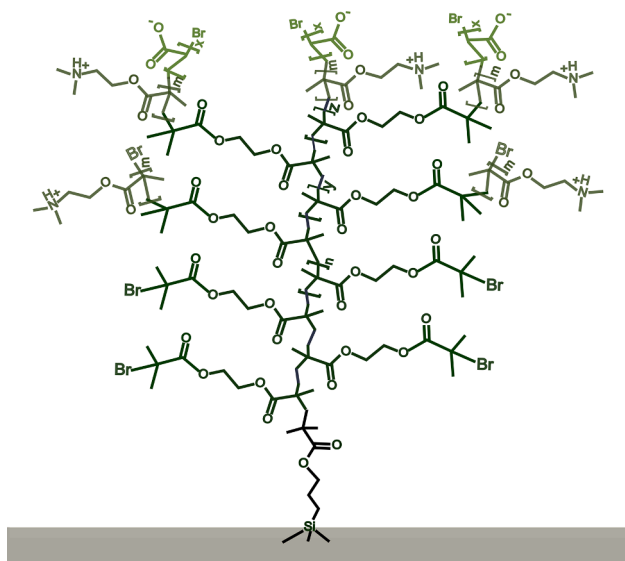


Fig. 1. AFM topography images with corresponding cross-section profiles generated for silicon wafer grafted with (a) PHEMA (Si-g-PHEMA), (b) PHEMA esterified by α -bromoisobutyryl bromide (Si-g-PHEMA-Br), complex architecture copolymer composed of PHEMA core and (c) PDMAEMA as branches (Si-g-PHEMA-g-PDMAEMA), (d) PDMAEMA and PtBA as block copolymer branches (Si-g-PHEMA-g-(PDMAEMA-b-PtBA)), (e) PDMAEMA and PAA as block copolymer branches (Si-g-PHEMA-g-(PDMAEMA-b-PAA)).



Scheme 2. Predicted structure of the polymer brushes with complex architecture generated using step-by-step SI-ATRP. The figure shows the brush core obtained by PHEMA bromination, which is present throughout the thickness of the polymer layer as well as the presence of PDMAEMA and PAA blocks mostly in the top of the coating.

calculated from GPC traces indicate linear correlation between the molar masses and monomer conversion (Fig. 2b) as well as present the increase of $M_{n,app}$ with ATRP time (Fig. 2c). In the case of dually-controlled photo-ATRP of DMAEMA the linear kinetic plot was observed (Fig. 2a), though the calculated initiation efficiency was determined as 124%, pointing out the trace occurrence of side reactions e.g. self-initiated radical polymerization of DMAEMA in solution. However, this competing reaction pathways occurring during PDMAEMA synthesis in water-based medium were described in more detail in our previous work [31]. Going further, in the case of water-immiscible monomer as *t*BA, it was necessary to select alcohol as a reaction medium. An alternative to the polymerization of that monomer might be ATRP in miniemulsion. However, this type of reaction mixture still remain a challenge to be applied for SI-ATRP [57] and could not be transferred to the microliter scale due to the need to conduct ATRP in miniemulsion under stirring conditions and elevated temperature [58].

Therefore, the ethanol was selected for SARA ATRP of *t*BA resulting in a well-defined polymer ($M_{n,app} = 39,400$ and $\bar{D} = 1.21$) (Fig. 2, d-f). The chemical structures of polymers formed in the solution were successfully confirmed via ^1H NMR techniques (see details in [Supplementary information, Fig. S6](#) and [Fig. S7](#)).

To create polyanion segments, Si wafers with attached polymer brushes were subjected to a specific hydrolysis under tetrafluoroacetic acid (TFA) conditions [59]. The deprotection of the *tert*-butyl groups to the carboxyl groups was verified by XPS and AFM measurements, which indicated a significant decrease in the dry film thickness of the brushes from non-homogeneous 244 nm (Fig. 1d) to 107 nm-thick polymer layer (Fig. 1e). It should be noted that for PtBA homopolymers, literature usually reports decrease in polymer layer thickness of 20–50% [13,60]. Therefore, taking into account the fact that PtBA represents only a part of the coating thickness (challenging to estimate), this decrease is unexpectedly large (compare Fig. 1d vs. Fig. 1e). Therefore, the observed change in thickness cannot be explained only by the loss of polymer mass due to removing of *tert*-butyl groups. It is supposed that the observed effect may be influenced by: (I) a change in the interactions between the polymer chains and their greater mutual compatibility, (II) possible side reactions with unknown causes e.g. occurring during hydrolysis (although hydrolysis of PDMAEMA chains was excluded by performing a test experiment, Fig. S8). As another reason (III), it should also be considered that each of the analyzed silicon wafers was placed separately in a common flask, which, despite the assumption of identical conditions, may lead to local layer thickness differences between the modified wafers (as can be observed in Fig. S5).

3.2. Si-g-PHEMA-g-(PDMAEMA-b-PAA) surface characteristic

At each stage of the polyelectrolyte coating synthesis, analysis of the wettability of the surface was carried out (Table S4, Fig. S9). The variations in the value of water contact angle correlating with changes in the structure of the macromolecules forming the hydrophilic film on the silicon wafer with $\text{WCA} = 31^\circ \pm 2^\circ$ for ampholytic nanolayer were observed (see details in [Supplementary data, chapter S2.5 WCA, SEM, EDS, XPS and ToF-SIMS coating analysis](#)). The free surface energy (FSE) was also determined on the basis of the WCA results (Table S5). As expected, the highest surface energy ($\gamma_s = 64.1 \pm 0.1$) was obtained for, Si-g-PHEMA-g-(PDMAEMA-b-PAA) due to its hydrophilic nature. Due to the multistep synthesis, the success of each stage was qualitatively confirmed using SEM imaging of the analyzed surface ([Supplementary information, Fig. S10](#) in chapter S2.5). Subsequently, scanning electron

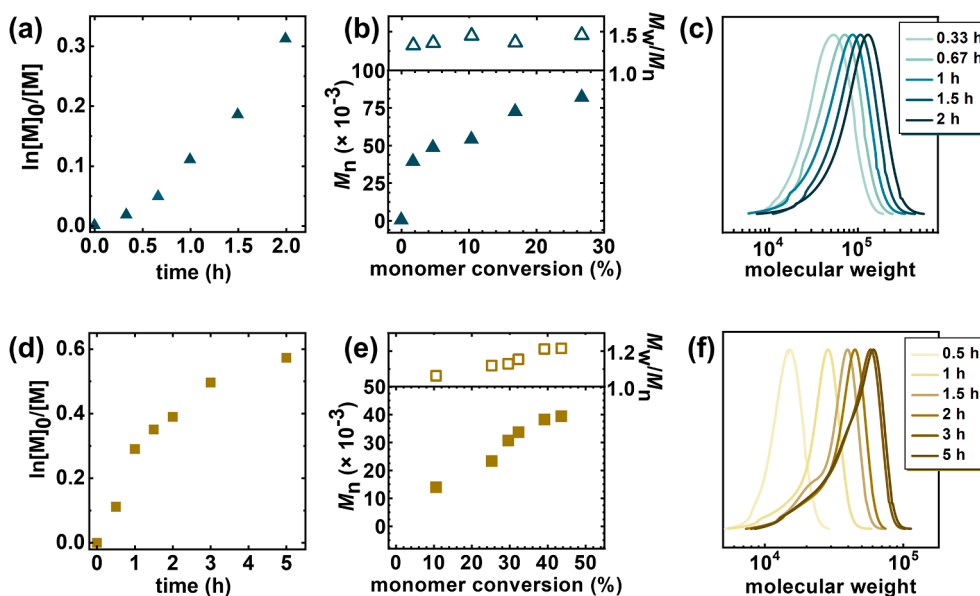


Fig. 2. (a) Semilogarithmic plot of DMAEMA conversion vs. photo-ATRP time; (b) M_n and M_w/M_n vs. DMAEMA conversion; (c) GPC traces of DMAEMA polymerization (Table 1, entry 2); (d) Semilogarithmic plot of tBA conversion vs. SARA ATRP time; (e) M_n and M_w/M_n vs. tBA conversion and (f) GPC traces of tBA polymerization (Table 1, entry 3).

microscopy with energy-dispersive X-ray spectroscopy (EDS) was used for further investigation to confirm the elemental composition of the modified silicon surface. Admittedly, EDS makes it possible to obtain results for the interior of the polymer layer (especially after gold sputtering), but it should be noted that with this technique (I) all light atoms have a higher detection threshold and (II) somewhat problematic is the small thickness of the layers which resulted in very low nitrogen detection for the samples studied. Nevertheless the elemental composition maps (Fig. S11) and the distribution of elements on the silicon wafer (Fig. S12) at each synthesis step (according to Scheme 1) were identified as expected. Importantly, after DMAEMA polymerization, a homogeneous distribution of nitrogen atoms was observed on the wafer surface, proving the addition of the PDMAEMA block to the brush architecture (Fig. S10c and Fig. S11c). Still, the disappearance of the nitrogen-derived signal for the surface treated with SARA ATRP of tBA (Fig. S10d) and hydrolysis of PtBA to PAA (Fig. S10e), might indicate grafting of the PtBA block as a copolymer substantially covering the PDMAEMA segment.

Furthermore, to confirm the effectiveness of the multi-step surface treatment procedure, the chemical composition of the polymer-coated Si wafer was studied by XPS (Table S7, Table 2). The presence of amine groups in the coating structure was also confirmed by XPS analyses, indicating the presence of C-N and C-N⁺ bonds (Fig. 3, Table 2). Deconvolution of the spectrum for sample Si-g-PHEMA-g-PDMAEMA

confirmed the presence of 284.8 eV (C-C), 285.4 eV (C-Br, C-C=O), 286.1 eV (C-N), 286.9 eV (C-O) and 288.9 eV (O-C=O) bonds (Table 2, Fig. 3) [8]. After deconvolution of the N1s spectrum, two lines were observed. The N1s line at a binding energy of 399.1 eV corresponds to a tertiary amine (N-C) and the line at 402.9 eV corresponds to a protonated amine (N⁺-C), which is semi-consistent with previous literature reports [61,62]. However, it is conjectured that the high content of C-N⁺ bonds can be assigned to partial quaternization of PDMAEMA brushes with alkyl bromides from the sacrificial ATRP initiator (EBiB) or parent Si-g-PHEMA-Br brushes having an excess of initiating groups. This phenomenon has not yet been well reported in the literature and will be verified in detail during further studies beyond the scope of this short report. Successively, considering the theoretical C/N ratio for PDMAEMA (where C:N = 8:1) and the value obtained for Si-g-PHEMA-g-PDMAEMA (where C:N = 10:1) suggesting the exposure of the PDMAEMA segment on the top of polymer coating. Similarly, in the case of the SARA ATRP of tBA, the XPS data for Si-g-PHEMA-g-(PDMAEMA-b-PtBA) result in C/O = 3:1 which is very close to the theoretical C/O = 3.5:1 value for PtBA. Such result indicates the exposure of bulky *tert*-butyl groups on the surface of the silicon wafer, resulting in the coverage of amide groups present in PDMAEMA segments proved by nitrogen level of 0% at (Table 2). Furthermore, the atomic percentages and corresponding binding energies of the signals C1s and O1s, clearly indicate a reduction in carbon content relative to oxygen after cleavage of the tBA

Table 2

XPS elemental analysis and contribution of the fitted peaks for Si wafers grafted with polymer brushes. Percentage of total atomic concentration of detected elements from analysis of two independent points, expressed as a mean $\pm$ std.dev. (% at.).

	C					N		O			Si	Br
	C-C	C-Br; C-C=O	C-N	C-O	O-C=O	N	N ⁺	O oxide	O-C	O=C		
Si-g-PHEMA-g-PDMAEMA	13.3 ± 0.7	17.1 ± 0.8	15.1 ± 1.2	14.0 $\pm$ 0.5	8.5 ± 0.1	2.7 ± 0.3	4.0 ± 0.3	3.3 ± 0.2	8.4 ± 0.1	12.4 ± 0.2	0.4 ± 0.0	0.8 ± 0.9
Si-g-PHEMA-g-(PDMAEMA-b-PtBA)	30.2	27.0	0.0	8.1	10.0	0.0	0.0	0.1	10.5	14.1	0.1	0.0
Si-g-PHEMA-g-(PDMAEMA-b-PAA)	± 0.4 31.2 $\pm$ 1.2	± 0.4 15.1	± 0.4 0.3	± 0.1 2.0	± 0.3 14.9	± 0.0 0.0	± 0.0 0.2	± 0.1 1.5	± 0.5 12.4	± 0.9 19.6	± 0.1 2.0	± 0.0 0.8
		± 1.3	± 0.4	± 0.7	± 0.3	± 0.0	± 0.3	± 0.7	± 0.3	± 0.9	± 0.6	± 0.1

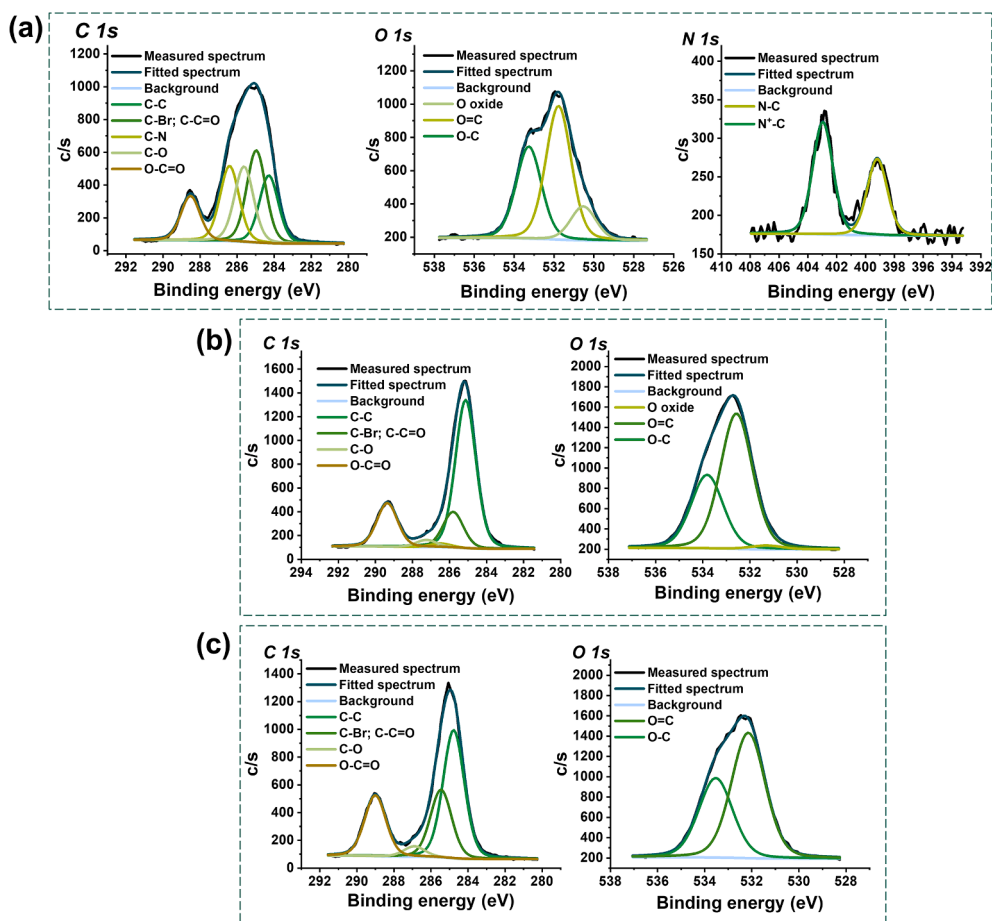


Fig. 3. XPS C1s, O1s, and N1s spectra with fitted lines for (a) Si-g-PHEMA-g-PDMAEMA, (b) Si-g-PHEMA-g-(PDMAEMA-*b*-PtBA) and (c) Si-g-PHEMA-g-(PDMAEMA-*b*-PAA).

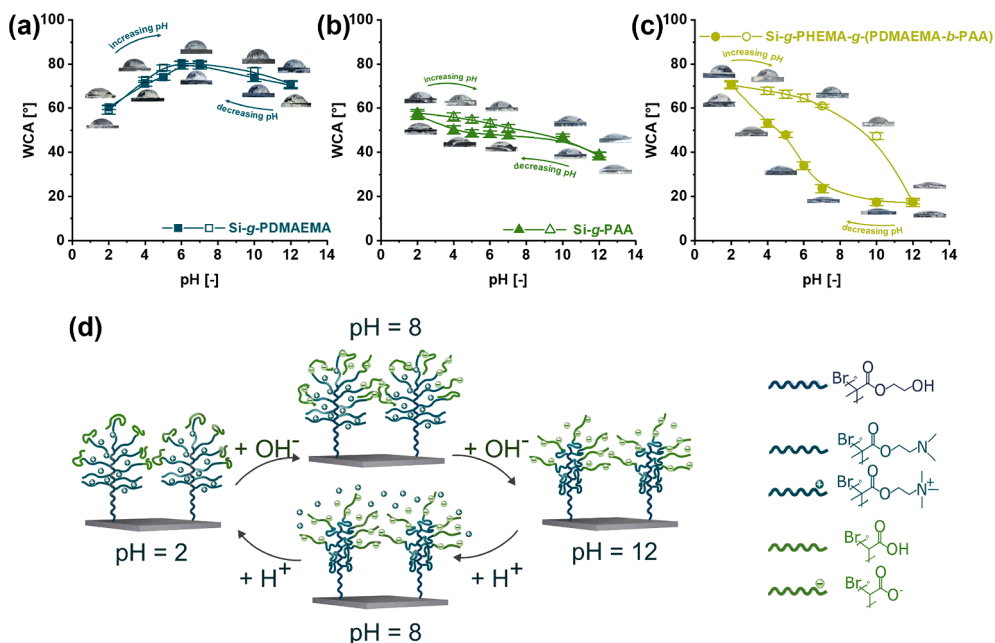


Fig. 4. Water contact angle measurement for Si wafers grafted with (a) PDMAEMA, (b) PAA and (c) PHEMA-g-(PDMAEMA-*b*-PAA) brushes in various pH. (d) Proposed behaviour of amphiphilic brushes in various pH.

groups. There was a decrease from 75% C1s for PtBA to 63% C1s for PAA and an increase of oxygen content from 23% O1s for PtBA to 34% O1s for PAA. This is consistent with the expected change in the chemical structure of the brush, and in line with previously published XPS characterization data for this type of compound [60]. It is a common observation that there is a strong desorption of halogens from the surface of organic compounds by X-rays. Therefore, it is not surprising that XPS measurements indicated only very low amount of bromine atoms in our brushes (Table S6, entry 1–2). XPS depth profiling tests were performed to iteratively access and determine the composition of the sub-surface layers in the final material (Fig. S13). The XPS profiling confirmed the presence of bromine throughout the thickness of the layer, and thus the retained functionality of the polymer chains, in accordance with the ATRP concept (Table S7). It is worth emphasizing that the highest concentration of bromine atoms was found near the brush/substrate interface confirming much lower participation of that initiating sites in grafting of PDMAEMA and PtBA segments.

Additionally, chemical composition of the PHEMA-g-(PDMAEMA-*b*-PAA) brush was verified using ToF-SIMS (Fig. S14). To this end three homopolymers, namely PHEMA, PDMAEMA and PtBA were examined and confirmed the presence of the peaks characteristic for each of the component in the spectrum collected for PHEMA-g-(PDMAEMA-*b*-PAA). What is more, the lack of $C_4H_9^+$ signal, originating from *tert*-butyl group, in positive ion ToF-SIMS spectrum of PHEMA-g-(PDMAEMA-*b*-PAA) brush proves successful hydrolysis of PtBA to PAA.

3.3. Study of pH-response of complex architecture polymer brushes

As PDMAEMA and PAA segments have different levels of ionization in deionized water depending on the solution in which they were previously incubated, it is possible to observe the hysteretic physico-chemical nature of the synthesized brush-based coating [63]. In order to explore these properties, the samples of homopolymer brushes and final surface-grafted PHEMA-g-(PDMAEMA-*b*-PAA) brushes were incubated for 30 min in unbuffered solutions of different pH to allow their ionization and adopting preferred conformation [64]. For Si-g-PAA at pH = 12, the wetting contact angle (WCA) was $\sim 39^\circ$ which is a typical observation for hydrophilic surfaces and is also due to the ionization of the vast majority of carboxyl groups present in the polyacrylic acid. In the case of Si-g-PHEMA-g-(PDMAEMA-*b*-PAA) WCA was even lower (17°) likely due to the presence of acrylic acid groups at the chain ends, an increased proportion of polyanion groups is assumed. For the same brushes at acidic conditions (pH = 2), a significantly higher WCA was observed 57° for Si-g-PAA and 70° Si-g-PHEMA-g-(PDMAEMA-*b*-PAA). This is an additional confirmation that PDMAEMA segments (that should be highly protonated and hydrophilic) are covered by the PAA segments that mostly determine the properties of most outer part of the brush layer. In the transition from acidic to basic pH, the wetting angle decreased which was associated with deprotonation and the return of the PAA to the hydrophilic state. Interestingly, the observed hysteresis often referred to as the memory effect for the copolymer coating was much more enhanced than for the homopolymer brushes (Fig. 4). The mechanism we propose is based on the pKa values for each polymer block – 4.5 for poly(acrylic acid) and 8.62 for poly(2-(dimethylamino) ethyl methacrylate). In pH = 2 both of them exist in acidic forms – PDMAEMA chains contain $-NH_3^+$ groups, whereas PAA $-COOH$ groups. As the pH increases, the carboxylic groups are becoming deprotonated to $-COO^-$, however, the amine groups are still present in cationic form, leading to strong ionic bonds between the countercharged polymers. After exceeding PDMAEMA pKa it drops its charge, leaving negatively charged PAA. Conversely, while lowering pH the high density of carboxylic groups prevent hydrogen cations from reaching PDMAEMA chains, thus enabling their straightening. Only after passing pH = 4.5, protonation of PAA results in its collapse and extending the PDMAEMA brushes, which explains the swift change in wettability between pH = 4 and pH = 6 in the Fig. 4c.

Subsequently, the ability of the ampholytic brushes on the Si surfaces to maintain pH-swelling behavior was evaluated via AFM by measuring thickness change of PHEMA-g-(PDMAEMA-*b*-PAA) layer (Fig. 5) immersed in solutions having acidic, neutral, and basic pH. According to wetting angle measurements (Fig. 4), and AFM measurements in liquids the obtained material demonstrate pH responsive behavior. In acidic media, the thickness of the brush layer almost tripled (h_{wet} in pH = 2 = 170 nm) due to the presence of the PDMAEMA segment relieving protonation at low pH, and a similar h_{wet} was recorded at pH = 6, which is partly the result of pKa of PDMAEMA in water equal to 8.62. However, it is admitted that the observed increase of layer thickness might be also caused by the swelling of non-ionized polymers. At the same time, the effect of PAA presence becomes apparent due to the pKa = 4.5 [65]. In the case of strongly alkaline conditions where pH = 10, the thickness of the polymer layer was more than five times higher than for dry films (h_{dry} = 62 nm vs. h_{wet} in pH=10 = 335 nm), which is related to swelling of the polymer chains caused by total deprotonation of the carboxyl groups and straightening of the PAA chains. It is important to underline that the determined swollen thicknesses of PHEMA-g-(PDMAEMA-*b*-PAA) are buried with some error related to 1) significant roughness of the coating and 2) substantial deformation of the swollen brushes even when using spherical probe during measurements.

3.4. μ L-Scale step-by-step synthesis of PHEMA-g-(PDMAEMA-*b*-PAA) brushes

Recently, diffusion-controlled polymerization and surface treatments at the microliter scale are one of the rapidly growing modifications of SI-ATRP approach [6,66]. Due to the increased efforts of the chemists to reduce the negative environmental impact of chemical processes in line with green chemistry trends, it is highly relevant to design processes that reduce the use of toxic chemicals and the amount of post-industrial waste [67,68]. Therefore, μ L-scale step-by-step synthesis of complex architecture polymers for limitation of process cost and waste reduction were designed as presented in Scheme 3.

In order to confirm the changes in the surface wettability occurring as a result of the sequential synthesis of polymer brushes, the water contact angle measurements were performed. The obtained values and observed changes were close to the data obtained for the samples modified using the microliter scale procedure (Fig. 6), thus confirming the successful modification and the legitimacy of minimizing the amount of reagents used for advanced surface treatments (more details in Supplementary Information Section S2.7). Additionally, visual verification of the sample was also carried out after each modification step (see photos in the Supplementary Information, Fig. S15). The observed color changes of the modified silicon wafers were as expected and correlated with the observation for film thickness recorded for samples modified at the mL-scale (compare with Fig. S5).

Depending on the process type and the composition of the reaction mixture, the use of a microlitre-scale system allowed a minimum of a tenfold reduction in costs, and in the case of the SI-ATRP of DMAEMA the process costs were reduced up to a 360-fold in comparison to the costs of the surface modification in a Schlenk flask (Table 3). The limiting factor in decreasing the overall cost of the synthesis turned out to be the concentration of the catalytic complex, which needs to be increased for most open-to-air systems as the presence of oxygen makes the ATRP more challenging. At the same time, the ligand – TPMA, being the part of the catalyst is the most high-priced reagent (\$506.00 for 5 g based on <https://www.tcichemicals.com>), which is the reason why decreasing its concentration to low ppm as well as reducing the volume of reaction mixture to microliters are profitable in both ecological and economic aspects. Important is also the fact that ligand should be used in excess to stabilize the complex. It was proved that 2-fold excess is sufficient, therefore using higher quantities e.g. 6-fold or 8-fold is unnecessarily increasing the overall costs [69]. However, reducing the amount and volume of catalyst still needs to be optimized, as procedures that

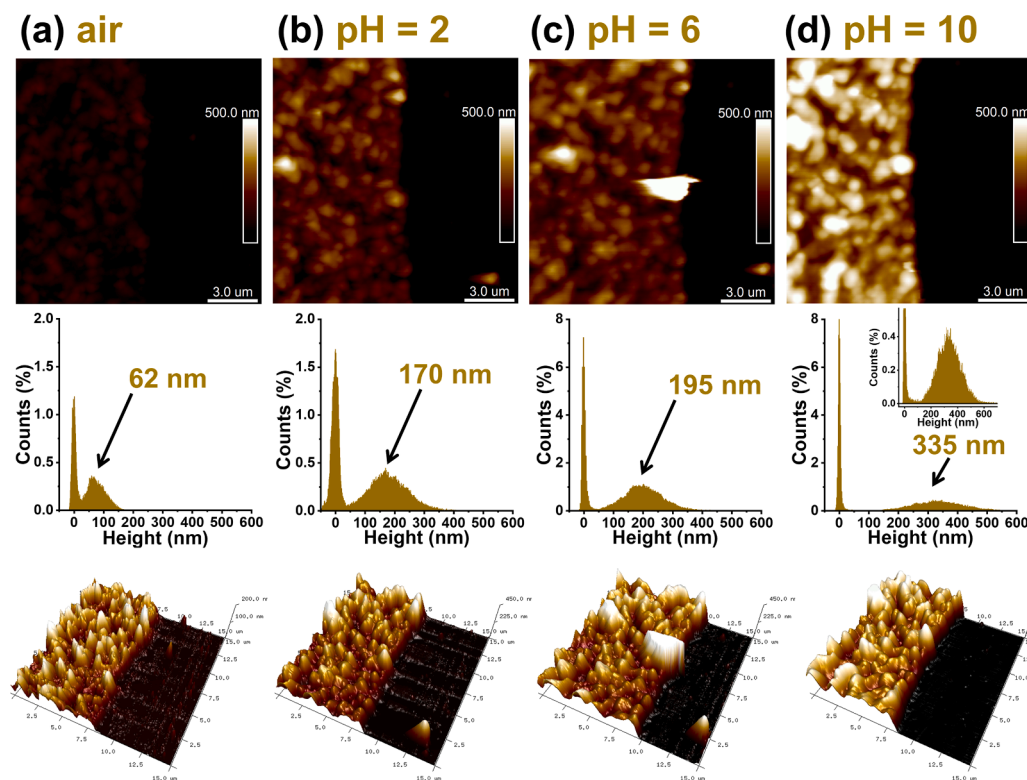


Fig. 5. AFM 2D and 3D topography images with corresponding depth histograms for Si-g-PHEMA-g-(PDMAEMA-b-PAA) analyzed in: a) air, b) pH = 2, c) pH = 6, d) pH = 10.



Scheme 3. Concept for the μL -scale step-by-step synthesis of PHEMA-g-(PDMAEMA-b-PAA) brushes. Abbreviations for reactions conditions: HEMA – 2-hydroxyethyl methacrylate, DMAEMA – 2-(dimethylamino)ethyl methacrylate and tBA – *tert*-butyl acrylate as monomers, EtOH – ethanol, NMP – *N*-methyl-2-pyrrolidone and DCM – dichloromethane as solvents, TFA – tetrafluoroacetic acid, BiBB – α -bromoisobutyryl bromide, Cu^0 – copper plate, $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ – catalytic complex, where TPMA – tris(2-pyridylmethyl)amine).

allow surface modifications of up to $\sim 40 \text{ m}^2$ under ambient conditions might require more than 2,000 ppm of complex [70].

Currently, one of the flagship green chemistry metrics describing the environmental impact of a chemical process is the E-factor, which defines the ratio of mass of the waste per mass of product [71]. In case of polymer synthesis in solution such as SARA ATRP of tBA (according Table 1, entry 3) the calculated environmental impact of waste (E-factor) for synthesis was equal 3.8 (and for the double-controlled photo-

ATRP of DMAEMA (according Table 1, entry 2) the value was determined as 2.8 (or 1.1 calculated without solvent), which are an adequate values for this type of process as for the bulk products the estimated E-factor was 1–5 [72]. Importantly, wastes should include only nonrecyclable postreaction compounds thereby the possibility of purification of final mixture to reuse the solvent (herein used ethanol as easily recyclable might be an example) or separation of catalyst complex after reaction to reuse might be the way to decrease the value of E-factor

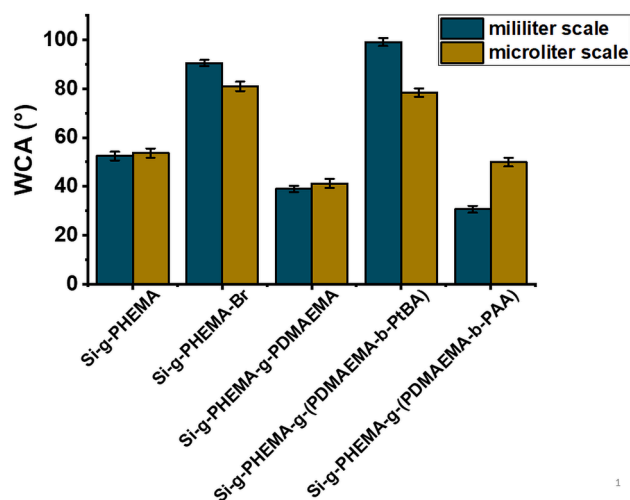


Fig. 6. WCAs of polymer coatings at each modification step synthesized according to μ L-scale and mL-scale procedures.

[73,74]. Furthermore, a theoretical approximation (based on monomer conversion determined by ^1H NMR) of the effect of $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ (catalyst complex) concentration on the E-Factor indicates that at 10 ppm the E-factor will be 2.8 and at 1,000 ppm it will be 3.2, justifying the need to use ATRP procedures with reduced catalyst complex concentration.

4. Conclusions

For the first time, a bottlebrushes-like ampholytic polymer coating was obtained using “green” sequential surface-initiated ATRP with a reduced concentration of the catalytic complex (less than 300 ppm by wt). The unique features of the ATRP enabled the synthesis of polymer brushes with a complex architecture, confirmed by a range of analytical techniques at each of the modification stages. Due to the breadth and variety of innovative aspects of the work, the most important observations summarizing the present paper were formulated as:

- (1) Considering high toxicity of traditionally used solvents such as DMF and high temperature in which the processes are conducted, replacing it with easy to recycle ethanol [75] or water:ethanol mixture, lowering the synthesis temperature to room temperature

and decreasing the concentration of catalyst to low ppm presents a big step forward in development of environmentally friendly procedures feasible to adapt to wide range of (meth)acrylates.

- (2) To present the versatility of our methods, a possibility of grafting branched polymer brushes, consisting of three types of polymer segments (core and two blocks of copolymers), from flat surfaces was demonstrated. The successful modifications were confirmed by detailed analysis concerning their composition, structure and properties.
- (3) The complex structure of the obtained polymer brushes led to conclusions indicating that in the case of PDMAEMA and PAA segments, the wettability properties of the material depend on pH, while being determined by the polymer exposed to the outer part of the coating. Moreover, the complex architecture of the brushes (Si-g-PHEMA-g-(PDMAEMA-b-PAA)) made it possible to observe much more intense hysteresis (memory effect) than in the case of polymer brushes built from a single type of mers (Si-g-PDMAEMA and Si-g-PAA, respectively).
- (4) Given the distinctive characteristics of weak polyelectrolytes, a structure consisting of polyanion and polycation blocks with divergent values of pK_a in the side chains, created the pH-responsive polymer brushes, able to react selectively with various compounds, therefore might be employed in drug delivery such as cyclodextrin for PDMAEMA [76] and bleomycin for PAA [11].
- (5) As high price, need for inert atmosphere and restriction concerning area of material are the main obstructions in industrial application of ATRP, reduction of solution consumed per cm^2 of modified surface from milliliter to microliter decreased the overall process cost by even 360 times. Additionally, the designed protocol enable grafting polymer brushes at ambient conditions and demonstrate the technique possible to easily scale up to semi-technical or technical scale.
- (6) Furthermore, the successful modification of silicon wafers playing the role of a model substrate suggests that this technique would be equally beneficial for other surfaces, like polyetheretherketone (PEEK), used in implantology, which high hydrophobicity is the main complication concerning implant-tissue integration. Hence, the highly hydrophilic polymer layer, containing groups accelerating the biomineralization process, might be the promising future solution.
- (7) Given the complex structure and applicability of optimized methods, this approach could undoubtedly be employed to monomers with other properties e.g. to synthesize

Table 3

Summary of reaction mixture cost depending to synthesis scale.

Reaction type	Volume of reaction mixture [$\mu\text{L}/\text{cm}^2$]		Reaction mixture volume reduction rate	Cost [$\$/\text{cm}^2$] ⁶		Cost reduction rate
	Milliliter	Microliter	Milliliter/Microliter	Milliliter	Microliter	Milliliter/Microliter
ARGET ATRP of HEMA ¹	7,200	20	360	1.042	0.006	170
esterification of PHEMA by BiBB ²	8,103	500	20	3.298	0.163	20
photo/ARGET ATRP of DMAEMA ³	7,200	20	360	1.415	0.004	360
SARA ATRP of tBA ⁴	7,200	20	360	1.380	0.010	133
hydrolysis of PrBA to PAA blocks ⁵	4,000	400	20	0.659	0.066	10

¹ Reaction conditions: **milliliter scale**: [HEMA]:[AsAc]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 1,000:2.5:0.25, EtOH as solvent, $t = 5$ h, $T = 60^\circ\text{C}$, inert atmosphere; **microliter scale**: [HEMA]:[AsAc]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 1,000:10:1, EtOH as solvent, $t = 5$ h, $T = \text{RT}$, ambient atmosphere, covered with glass plate to reduce oxygen diffusion.

² Reaction conditions: **milliliter and microliter scale**: $V_{\text{BiBB}}:V_{\text{NMP}} = 1.05:3$, $t = 48$ h, $T = \text{RT}$, ambient atmosphere.

³ Reaction conditions: **milliliter scale**: [DMAEMA]:[EBiB]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 2,400:1:0.72, EtOH:H₂O = 2:3 as solvent, $t = 2$ h, $T = \text{RT}$, inert atmosphere, blue light with $\lambda_{\text{max}} = 460$ nm was used as reducing factor; **microliter scale**: [DMAEMA]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 2,400:0.72, EtOH:H₂O = 2:3 as solvent, $t = 2$ h, $T = \text{RT}$, ambient atmosphere, covered with glass plate to reduce oxygen diffusion, no irradiation – monomer as the only reducing factor.

⁴ Reaction conditions: **milliliter scale**: [tBA]:[EBiB]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 1,000:1:0.3, EtOH as solvent, $t = 5$ h, $T = 60^\circ\text{C}$, inert atmosphere, copper wire was used as reducing factor ($S/V = 0.1$); **microliter scale**: [tBA]:[$\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$] = 1,000:2, EtOH as solvent, $t = 24$ h, $T = \text{RT}$, ambient atmosphere, covered with copper plate acting as reducing agent and reducing oxygen diffusion.

⁵ Reaction conditions: **milliliter and microliter scale**: 10% solution of TFA in DCM, $t = 24$ h, $T = \text{RT}$, ambient conditions.

⁶ Approximate cost of process was calculated basing on prices of chemical compounds adopted from Sigma Aldrich website and their volumes consumed in the process.

superhydrophobic, antimicrobial or flame-retardant hybrid materials with enhanced activity due to higher than usual density of functional groups.

CRediT authorship contribution statement

Monika Flejszar: Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kinga Ślusarczyk:** Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Agata Hochół:** Investigation, Data curation, Writing – original draft. **Paweł Chmielarz:** Conceptualization, Methodology, Validation, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Magdalena Wytrwal:** Investigation, Data curation, Writing – original draft, Visualization. **Karol Wolski:** Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Kaja Spilarewicz:** Investigation, Data curation, Writing – original draft, Visualization. **Kamil Awiśuk:** Investigation, Data curation, Writing – original draft, Visualization. **Joanna Raczkowska:** Investigation, Data curation, Writing – original draft, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eurpolymj.2023.112142>.

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Sequential SI-ATRP in μ L-scale for surface nanoengineering: A new concept for designing polyelectrolyte nanolayers formed by complex architecture polymers

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S1. Materials and methods

S1.1 Materials

Polished silicon wafers were purchased from ON Semiconductor (Czech Republic). 3-(trichlorosilyl)propyl 2-bromo-2-methylpropanoate was purchased from TCI Chemicals. 2-bromoisobutyryl bromide (BiBB, 98%), ethyl 2-bromoisobutyrate (EBiB, 98%), triethylamine (TEA, >99.5%), tetrahydrofuran (THF, >99.9%), acetone (DMK, >99.9%), deuterated chloroform (CDCl_3 , >99.8%), trifluoroacetic acid (TFA, >99%), ascorbic acid (AA, >99%) and sodium bromide (NaBr, >99%) were purchased from Sigma Aldrich. Dichloromethane (DCM, >99.5%), toluene (TOL, >99.5%), methanol (MeOH, >99.8), ethanol (EtOH, 99.9%), hydrogen peroxide (H_2O_2 , 30%), sulfuric acid (SAC, >95%) and anhydrous magnesium sulphate (MgSO_4 , 98.5%) were purchased from Chempur. *N,N*-Dimethylformamide (DMF, 99.9%) and lithium chloride (LiCl , >99%) were purchased from Acros. Deuterium oxide (D_2O , 99.9%) were purchased from Deutero. These reagents were used without further purification. Tris(2-pyridylmethyl)amine (TPMA) was synthesized in our research group according to published procedure [1]. $\text{Cu}^{\text{II}}\text{Br}_2$ /TPMA stock solutions were prepared according to reference [2]. 2-(Dimethylamino)ethyl methacrylate (DMAEMA, >99%), *tert*-butyl acrylate (*t*BA, >99%), 2-hydroxyethylmethacrylate (HEMA, >99%), were purchased from Sigma Aldrich and passed through a column filled with basic alumina prior to use to remove monomethyl ether hydroquinone as inhibitor [3]. Cu^0 wire was purchased from Alfa Aesar (99.9%).

S1.2 Analysis

^1H NMR spectra in D_2O and CDCl_3 were measured using Bruker Avance 500 MHz spectrometer. Monomer conversion and theoretical number-average molecular weight ($M_{n,\text{th}}$) were determined by ^1H NMR [4, 5].

GPC analysis of polymers generated in solution was performed using a Shimadzu modular system consisting of a CBM-40 system controller, RID-20A differential refractive-

index detector a SIL-20AHT automatic injector with two Repro-Gel GPC 5 μm columns, pore size: 10^4 Å and 10^5 Å, thermostated at 30°C by a CTO-20A oven. The eluent in chromatographic separation were tetrahydrofuran *N,N*-dimethylformamide (HPLC grade with 0.01% w/v LiCl) with the flow rate set at 1 mL min⁻¹. A molecular weight calibration curve were determined based on monodispersed polystyrene standards (PSS, Polymer Standards Service). The apparent molecular weights (MWs) and molecular weights distributions (MWD, *D*) were determined using GPC Postran Analysis software provided by Shimadzu Corporation.

AFM measurements were carried out using a Dimension Icon AFM microscope (Bruker, Santa Barbara, CA, USA) working in the PeakForce Tapping (PFT) and QNM® mode. AFM images with a total resolution of 256×256 pixels were captured at the edge of the scratch made by metal tweezers or needle. Before the measurement, the scratched brushes were washed by organic solvents such as ethanol and toluene and the thicknesses were determined from the depth histograms. 5×5 μm topography images of the obtained layers were captured with higher resolution (384×384 pixels) to better visualize the morphology of the coatings. AFM measurements in air were performed using standard ScanAssyst probes (Bruker, Santa Barbara, CA, USA) with nominal spring constant of 0.4 N/m (nominal tip radius 2-5 nm), while measurements in liquid using pre-calibrated and spherical biosphere B300-FM probes (calibrated spring constant 5.2 N/m, tip radius 300 nm, Nanotools, München, Germany). AFM measurements in liquids were performed under minimal loading force (typically 2nN) to minimize the deformation of the swollen brushes.

The elemental composition and distribution of complex architecture polymers grafted from the surface were determined by energy-dispersive X-ray spectroscopy (EDS) with the use of an Oxford Instruments X-Act EDS detector and quantified on the basis of calibration standards. A surface fragment was scanned at least 15 times to obtain elemental maps. Measurements were conducted using a low beam energy (5 kV). Scanning electron microscopy

(Tescan Vega SEM) images of the mapped surface were also collected using SE and BSE detectors. High-resolution images of the surface were performed at 30 kV. Before SEM analysis, the samples were coated with a thin layer of gold,

Polymeric surfaces were analysed with a TOF.SIMS 5 instrument (ION-TOF GmbH) using a 30 keV bismuth liquid metal ion gun and Bi_3^+ clusters. The surface spectra were recorded from different non-overlapping $200\text{ }\mu\text{m} \times 200\text{ }\mu\text{m}$ regions with a high mass resolution of $m/\Delta m > 6600$ at C_4H_5^+ ($m/z = 53$). To ensure a static mode condition, the dose density deposited on the surface was less than 10^{12} ions/ cm^2 for all the measure points.

Elemental analysis of samples was performed using scanning X-ray photoelectron spectrometer a PHI 5000 Versa Probe II (ULVAC-PHI, Chigasaki, Japan) with a micro-focused ($100\text{ }\mu\text{m}$, 25 W) Al $\text{K}\alpha$ X-ray beam with photoelectron take-off angles of 45° . Photoelectron spectra were collected at two independent areas for each $300\text{ }\mu\text{m} \times 300\text{ }\mu\text{m}$ sample, the working pressure of the analytical chamber was less than 5×10^{-7} , the analyzer pass energy was 46.95 eV. The charge-up effect was compensated by using a dual-beam charge neutralizer. All XPS peaks were referenced to the C1s C-C line at a binding energy of 284.8 eV. The PHI MultiPak software (version 9.9.2) was used to deconvolute the spectra, and the Shirley method was used to subtract the spectrum background. To characterize composition of complex architecture polymers as a function of penetration depth in the sample, XPS depth-profiling was performed using an argon gas cluster ion beam (Ar-GCIB). The average Ar cluster size was around 4000 atoms. The beam energy was set to 10 keV, the beam current was 30 nA, 21 cycles, 8 mins sputtering per cycle, and the sputtered area was $3 \times 3\text{ mm}^2$. Measurements were taken from a $300\text{ }\mu\text{m} \times 300\text{ }\mu\text{m}$ area with a Zalar rotation to standardize the surface.

Contact angle measurements (Θ , WCA) of the grafted polyelectrolyte brushes were performed after each synthesis step using liquids of different polarity (water, formamide and diiodomethane). In addition, WCA values at different pH (2, 4, 5, 6, 7, 10, 12) were measured

for the obtained ampholytic coatings (Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA)) and for the homopolymer Si-*g*-PDMAEMA and Si-*g*-PAA brushes. The plates were incubated for 30 minutes before each measurement at the specific pH, which was adjusted using 0.1 M HCl or 0.1 M NaOH solutions. Between the measurements, the plates were blown with air. Two series of measurements were performed – at first using solutions in decreasing pH manner and then in increasing pH manner to determine the difference in brushes behavior resulting in hysteresis. All measurements were performed at room temperature. Photos of 2 μ L droplets of various liquids, placed onto the modified surfaces, were taken using a Panasonic DC-FZ82 camera equipped with LUMIX DC VARIO 1:2.8-5.9/3.58-215 ASPH lens and analysed by Kropla software [6]. For each liquid, the results are calculated using the arithmetic mean of five drops. Based on the obtained WCA values, the surface free energy (FSE) was calculated according to the Owens-Wendt method using the Energia software [7].

S1.3 Procedures

S1.3.1 Immobilization of the initiator on silicon wafer surface

Immobilization of the initiator on the surface of the silicon wafer was carried out according to the following procedure [8]. First, the silicon substrates were ultrasonicated in ethanol for 10 minutes and then cleaned using a UV ozonator with a UV lamp with dominant wavelengths: 185 and 254 nm (Ossila, Sheffield, UK). Subsequently, the trichlorosilane initiator was immobilized according to the following procedure [9]. Purified substrates were immersed in a toluene solution containing 3-(trichlorosilyl)propyl 2-bromo-2-methylpropanoate (1 mM) and triethylamine (10 mM) and left under an argon atmosphere for 24 h. After that, the silicon wafers were rinsed with toluene, ultrasonicated in toluene for 1 minute and dried in an argon stream.

S1.3.2 Polymerization of HEMA via low ppm ATRP

Milliliter-scale synthesis via SARA ATRP

5 mL (0.041 mol) of HEMA, 4.75 mL of DMF, 246.7 μL (12.3 μmol , 300 ppm) of $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ stock solution (0.05 M in DMF), 6.03 μL (40 μmol) of EBiB and magnetic stirring bar was introduced into the Schlenk flask. The reaction flask was closed with a rubber septum equipped with Cu^0 wire ($l = 12.7$ cm, $d = 1$ mm) previously activated with HCl and washed with THF. The reaction mixture was degassed with argon for 30 minutes, then Cu^0 wire was dropped into Schlenk flask to start the polymerization. Synthesis of PHEMA was carried out for 3 h at 60°C with a vigorous stirring rate (900 rpm).

Milliliter-scale synthesis via ARGET ATRP

Synthesis of PHEMA was carried out for 5h at 60°C with a vigorous stirring rate (900 rpm). Reaction mixture prepared from HEMA (9.0 mL, 0.074 mol), ethanol (7.978 mL), $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ (370 μL , 18.5 μmol , 250 ppm) stock solution (0.05 M in DMF). The reaction flask equipped with a magnetic stirring bar was closed with a rubber septum with the cable ties containing silicon wafers with initiator and degassed with argon for 1h. . Polymerization was launched by adding ascorbic acid (651.6 μL , 0.185 mmol) as a reducing agent (stock solution 50 mg/mL).

Microliter-scale synthesis

72.4 μL , 0.021 mmol of ascorbic acid (stock solution 50 mg/ml in H_2O) was added to a solution consisting of HEMA (0.25 mL, 2.1 mmol), EtOH (0.136 mL 2.3 mmol), $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ (41.1 μL , 20.5 μmol , 1000 ppm) stock solution (0.05 M in DMF) and the 10 μL of mixture was quickly applied with an automatic pipette on a silicon wafer placed on a glass slide. The silicon wafer with the reaction mixture was covered with glass slides to minimize oxygen access and left for 6 hours under ambient conditions. The volume ratio of the reaction mixture (V) to the surface of the silicon wafer (S) was 20 $\mu\text{L}/\text{cm}^2$.

S1.3.3 Synthesis of PHEMA-Br macroinitiator

Milliliter-scale synthesis

10 mL of NMP was added to the Schlenk flask equipped with a magnetic stirring bar, sealed with a septum fitted with cable ties containing silicon wafers grafted with PHEMA and placed in ice bath. BriBBBr (5.257 mL) was dissolved in 5 mL NMP and was dripped into the sealed flask for 45 min. Synthesis was then conducted with a vigorous stirring (900 rpm) for 4 days in room temperature. The prepared macroinitiator was used for DMAEMA polymerization.

Microliter-scale synthesis

Silicon wafers grafted with PHEMA were placed in a small beaker, 740.5 μL of NMP was added and the beaker was sealed with a rubber septum. Then 260 μL of BriBBBr was injected. The beaker was left at room temperature for three days.

S1.3.4 Polymerization of DMAEMA via dually-controlled photoATRP

Milliliter-scale synthesis

DMAEMA was filtered through the syringe with filter with pore size 0.2 μm and basic Al_2O_3 in order to remove the monomethyl ether hydroquinone as inhibitor. Synthesis of PDMAEMA was conducted with a vigorous stirring rate (900 rpm) in room temperature. 9 mL (0.054 mol) of DMAEMA was introduced into a glass vial and tightly plugged with a rubber septa. 3.6 mL of EtOH, 5.072 mL of distilled water, 3.30 μL (20 μmol) of EBiB and 324.3 μL (16.2 μmol , 300 ppm) of $\text{Cu}^{\text{II}}\text{Br}_2$ /TPMA stock solution (0.05 M in DMF) were added to a Schlenk flask equipped with a magnetic stirring bar. The reaction flask was closed with a rubber septum equipped with the cable ties containing silicon wafers with PHEMA-Br macroinitiator. The sealed flask and glass vial were degassed with argon for 1 hour. Thereafter DMAEMA was injected into the Schlenk flask and a blue LED-powered photoreactor was switched on to start the reaction (emitting light with $\lambda_{\text{max}} = 460 \text{ nm}$). Samples of the reaction

mixture were withdrawn periodically to follow the monomer conversion using ^1H NMR in D_2O . The M_n and MWD were determined by GPC measurements using DMF + 50 mM LiCl as eluent.

Microliter-scale synthesis

DMAEMA was filtered through the syringe with filter with pore size $0.2\ \mu\text{m}$ and basic Al_2O_3 in order to remove the monomethyl ether hydroquinone as inhibitor. A $10\ \mu\text{L}$ of solution, consisting of $0.25\ \text{mL}$ of DMAEMA ($2\ \text{mmol}$), $0.1\ \text{mL}$ of EtOH, $0.141\ \text{mL}$ of H_2O (EtOH/ H_2O ratio was 40/60), $9.0\ \mu\text{L}$ ($45\ \mu\text{mol}$, $300\ \text{ppm}$) of $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$, was applied on a silicon wafer (Si-*g*-PHEMA-Br) placed on a glass slide. To minimize oxygen access, glass slides were placed over the silicon wafer with the reaction mixture. The wafer was left at ambient conditions for two hours. The reaction system was not irradiated, a monomer was used as reducing agent according to the procedure [8].

S1.3.5 Polymerization of *t*BA via SARA ATRP

Milliliter-scale synthesis

Synthesis of PtBA was conducted with a vigorous stirring rate ($900\ \text{rpm}$) in 60°C . Ethanol ($4.31\ \text{mL}$, $0.056\ \text{mol}$), *t*BA ($4.50\ \text{mL}$, $0.031\ \text{mol}$), $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ ($185.2\ \mu\text{L}$, $9.26\ \mu\text{mol}$, $300\ \text{ppm}$) stock solution ($0.05\ \text{M}$ in DMF), EBiB ($4.53\ \mu\text{L}$, $0.031\ \text{mmol}$) were added to a Schlenk flask equipped with a magnetic stirring bar. The reaction flask was closed with a rubber septum equipped with the cable ties containing silicon wafers (Si-*g*-PHEMA-*g*-PDMAEMA-Br) and Cu^0 wire (dimension = $2.8\ \text{cm}$ length; thickness $1\ \text{mm}$, $S/V = 0.1$). In the previous step, copper wire was activated with HCl and washed with THF. After 30 minutes of degassing with argon, the reaction was initiated by adding Cu^0 wire to the reaction flask. Samples for ^1H NMR and GPC analysis were periodically taken using nitrogen sparged needles. In order to remove the catalyst, polymer samples were dissolved in THF and toluene was added as an external standard-mobile phase, then passed through a neutral alumina column with $0.22\ \mu\text{m}$ syringe filter before GPC analysis.

Microliter-scale synthesis

A 10 μL of reaction mixture, consisting of *t*BA (0.25 mL, 2 mmol), EtOH (0.181 mL) and $\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}$ (68.6 μL , 34.3 μmol , 2000 ppm), was applied on a silicon wafer (Si-*g*-PHEMA-*g*-PDMAEMA-Br) placed on a glass slide. The silicon wafer with the reaction mixture was covered with Cu plate at a distance of about 50 μm . The reaction was carried out for 24 hours under ambient conditions.

S1.3.6 Acidic hydrolysis of PtBA to PAA

Milliliter-scale synthesis

Silicon wafer (Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PtBA)) was placed in a 10 mL glass vial. 1.8 mL of dichloromethane was added, and the vials were sealed with rubber septa and cable ties. Then 200 μL of trifluoroacetic acid was injected. The vial was left at room temperature for 24 hours.

Microliter-scale synthesis

Silicon wafer (Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PtBA)) was placed in a 10 mL glass vial and 180 μL of dichloromethane was added. The vial was sealed with rubber septa and cable tie, then 20 μL of trifluoroacetic acid was injected. The vial was left at room temperature for 24 hours.

PDMAEMA hydrolysis test

The PDMAEMA homopolymer was dissolved in DCM, after which 10% vol trifluoroacetic acid was added. The mixture was stirred at room temperature for 24 h. After that the product was purified by dialysis in THF and the solvent was removed in rotary evaporator. To verify the changes in structure ATR-IR spectra were performed.

S1.3.7 Purification of silicon wafers

A surfaces were cleaned in an ultrasonic cleaner in order to remove physical absorbed contaminants and residues of the reaction mixture. Directly at the end of each reaction, the silicon wafers were washed with ethanol or THF and then sonicated in ethanol or THF two times for 15 minutes, dried under air stream and then characterized using AFM [10].

S2. Results and discussion

S2.1 Bromination of silicon wafers

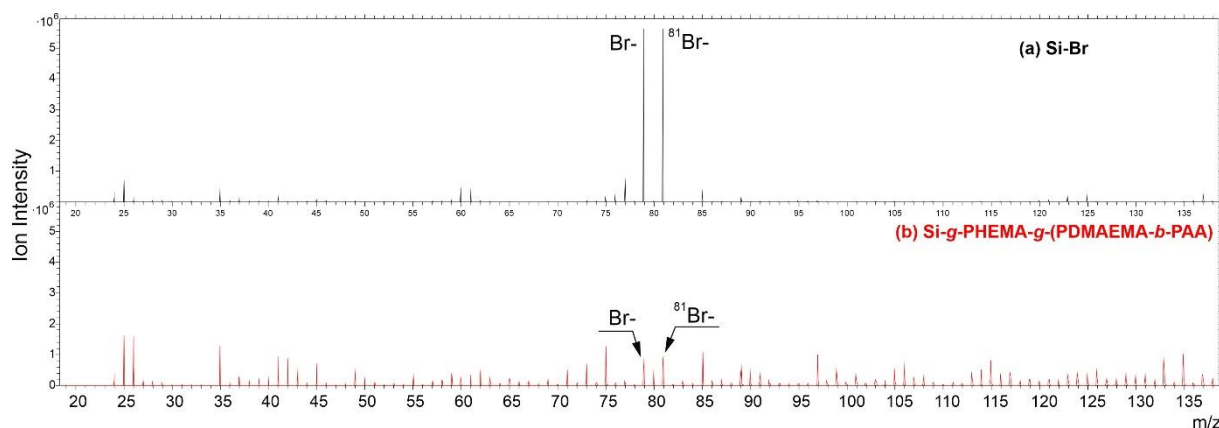


Figure S1. Representative negative ions ToF-SIMS spectra of Br modified Si substrate (a) and PHEMA-*g*-(PDMAEMA-*b*-PAA) bottlebrush-like polymers (b)

S2.2 Optimization of PHEMA synthesis conditions

The choice of ATRP method and solvent for polymerization of HEMA started with the SARA ATRP in DMF (**Table S1**, entry 1). However, high dispersity of the final product obtained was unsatisfactory compared to the analogous polymerization performed according to the ARGET ATRP concept. Subsequently, the ARGET ATRP with 2-fold excess of ascorbic acid as reducing factor ($[AA]:[Cu^{II}Br_2/TPMA] = 2$) was put on test. This approach was verified with fivefold and 20-fold excess of ascorbic acid, respectively. However, the 20-fold excess of vitamin C was found to be too high, as the reaction under these conditions was characterized by too high initiation efficiency = 200%, indicating the presence of undesired side reactions. In a next step, ARGET ATRP in ethanol was tested in order to use a “greener” solvent than DMF. Both performed reactions were controlled in time and resulted in polymer characterized by narrow molecular mass distribution (**Figure S2**). Finally, a 10-fold excess of reducer ($[AA]:[Cu^{II}Br_2/TPMA] = 10$) was chosen as optimal, with monomer conversion = 48 %, high agreement between $M_{n,theo}$ and $M_{n,app}$, and initiation efficiency = 103.1%. Therefore, ARGET

ATRP in ethanol with a 10-fold excess of ascorbic acid was used to polymerize HEMA *via* low ppm ATRP.

Table S1. Optimization of PHEMA synthesis conditions in solution.

Entry	[AA]:[complex]	Reaction time [h]	Solvent	Conv ² [%]	k_p^{app} ² [h ⁻¹]	$M_{n,theo}$ ³	$M_{n,app}$ ⁴	$\bar{D}$ ⁴	I_{eff} ⁵ [%]
1 ¹ (4)	-	3	DMF	37	0.18	48,800	48,700	1.73	100.2
2 (6)	2	2	DMF	13	- ⁶	17,200	27,500	1.13	62.5
3 (7)	5	2	DMF	19	- ⁶	25,500	24,000	1.18	106.3
4 (8)	20	2	DMF	47	- ⁶	61,700	29,500	1.23	208.8
5 (21)	10	1	EtOH	48	- ⁶	62,600	60,700	1.35	103.1
6 (22)	2	0,25	EtOH	8	- ⁶	10,300	25,300	1.17	40.7

General reaction conditions: T = 60 °C (except for entry 5 where T = rt); V_{tot} = 6 mL (except for entry 1); [I]₀ = 4.1 mM; [HEMA]₀ = 4.11 M; DP_{target} = 1,000; [Cu^{II}Br₂/TPMA]₀ = 300 ppm. 2-Fold excess of TPMA was used to stabilize the complex. ¹SARA ATRP: V_{tot} = 10 mL; Cu⁰ wire with a ratio of surface area (S) to total volume of reaction mixture (V), S/V = 0.4 cm⁻¹ (l = 12.7 cm, d = 1 mm). ²Monomer conversion (Conv), theoretical M_n and apparent propagation rate coefficient (k_p^{app}) were determined by ¹H NMR.; ³ $M_{n,th} = ([M]_0/[I]_0) \times Conv \times M_{monomer} + M_{initiator}$; ⁴ $M_{n,app}$ and $\bar{D}$ was determined by DMF+50mM LiCl GPC with PS standards. ⁵Initiation efficiency: $I_{eff} = (M_{n,theo}/M_{n,app}) \cdot 100\%$. ⁶ The first and last samples were taken for ¹H NMR to calculate the conversion, reaction kinetics were not investigated.

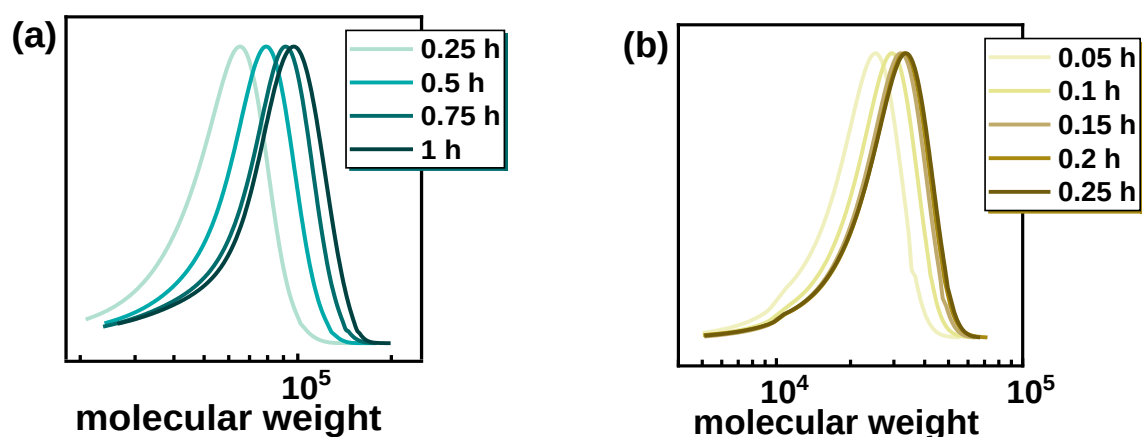


Figure S2. GPC traces of PHEMA chains prepared in EtOH according to **Table S1** a) entry 5, b) entry 6, using DMF + 50 mM LiCl as eluent.

In the next step, the proposed technique was applied to modify silicon wafers' surface. Tests of surface-initiated ARGET ATRP with and without the addition of a free initiator suggested the possibility of synthesizing a thicker PHEMA layer on silicon wafers in case of reactions without a sacrificial initiator, which consumes part of the monomer present in the polymerization mixture to form macromolecules in solution. Moreover, in this case the addition of the sacrificial initiator to polymerization mixture resulted in the synthesis of inhomogeneous PHEMA brushes, with additionally locally adhered homopolymer, difficult to wash off from the silicon wafer. In the next step, the reaction time (**Table S2**, entry 3 and 4) and the concentration of the catalytic complex (**Table S2**, entry 3 and 5) were verified.

Table S2. Optimization of grafting PHEMA brushes from Si wafers.

Entry ¹	[Cu ^{II} Br ₂ /TPMA] ₀ [ppm]	Reaction time [h]	Conv ² [%]	k_p^{app} ² [h ⁻¹]	$M_{n,theo}$ ³	$M_{n,app}$ ⁴	$\bar{D}$ ⁴	Silicon wafers images
1 ⁵ (27)	500	3	60	-	78,900	71,200	1.37	
2 ⁶ (28)	500	5	-	-	-	-	-	

3⁶ (29)	500	5	-	-	-	-	-	
4⁶ (30)	500	3	-	-	-	-	-	
5⁶ (31)	250	5	-	-	-	-	-	

¹General reaction conditions: T = 60 °C; V_{tot} = 6 mL; solvent: EtOH (for entry 2: DMF); [I]₀ = 4.1 mM for entry 1; [DMAEMA]₀ = 4.11 M; DP_{target} = 1,000; [AA]:[complex] = 10. 2-Fold excess of TPMA was used to stabilize the complex. ²Monomer conversion (Conv), theoretical M_n and apparent propagation rate coefficient (k_p^{app}) were determined by ¹H NMR. k_p^{app} was determined as a slope of the line $\ln[M]_0/[M] = f(t)$; ³ $M_{n,th} = ([M]_0/[I]_0) \times \text{Conv} \times M_{monomer} + M_{initiator}$; ⁴ $M_{n,app}$ and $\bar{D}$ was determined by DMF+50mM LiCl GPC with PS standards. ⁵Gel. ⁶Synthesis without sacrificial initiator, therefore neither conversion nor $M_{n,th}$, $M_{n,app}$ and $\bar{D}$ were determined, reaction kinetics were not investigated.

S.2.3 Optimization of PtBA synthesis conditions

Table S3. Optimization of PtBA synthesis conditions in solution

Entry	$[\text{Cu}^{\text{II}}\text{Br}_2/\text{TPMA}]_0$ [ppm]	Cu^0 S/V ¹	Conv ² [%]	k_p^{app} ² [h ⁻¹]	$M_{n,\text{theo}}$ ³	$M_{n,\text{app}}$ ⁴	$\bar{D}$ ⁴	I_{eff} ⁵ [%]
1 (11)	300	0.4	85	- ⁶	109,400	60,800	1.27	179.9
2 (12)	100	0.4	62	- ⁶	80,000	33,300	1.25	242.3
3 (13)	300	0.1	65	- ⁶	83,200	46,800	1.18	177.8

General reaction conditions: T = 60 °C; V_{tot} = 6 mL; t = 2 h; solvent: DMF; [I]₀ = 3.4 mM; [tBA]₀ = 3.43 M; DP_{target} = 1,000. 2-Fold excess of TPMA was used to stabilize the complex. ¹Cu⁰ wire with a ratio of surface area (S) to total volume of reaction mixture (V) (*d* = 1 mm, *l* = 7.6 cm for entry 1 and 2, *l* = 1.86 cm for entry 3). ²Monomer conversion (Conv) and theoretical *M_n* were determined by ¹H NMR.; ³ $M_{n,\text{th}} = ([\text{M}]_0/[\text{I}]_0) \times \text{Conv} \times M_{\text{monomer}} + M_{\text{initiator}}$; ⁴ $M_{n,\text{app}}$ and $\bar{D}$ was determined by DMF+50mM LiCl GPC with PS standards. ⁵Initiation efficiency: $I_{\text{eff}} = (M_{n,\text{theo}}/M_{n,\text{app}}) \cdot 100\%$. ⁶The first and last samples were taken for ¹H NMR to calculate the conversion, reaction kinetics were not investigated.

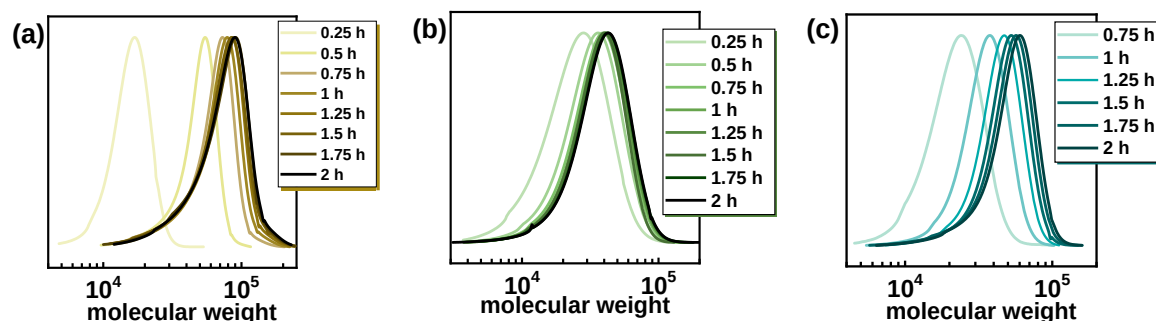


Figure S3. GPC traces of PtBA chains prepared according to **Table S3** a) entry 1, b) entry 2, c) entry 3 using DMF + 50 mM LiCl as eluent.

S.2.4 Step-by-step synthesis of complex architecture polymers

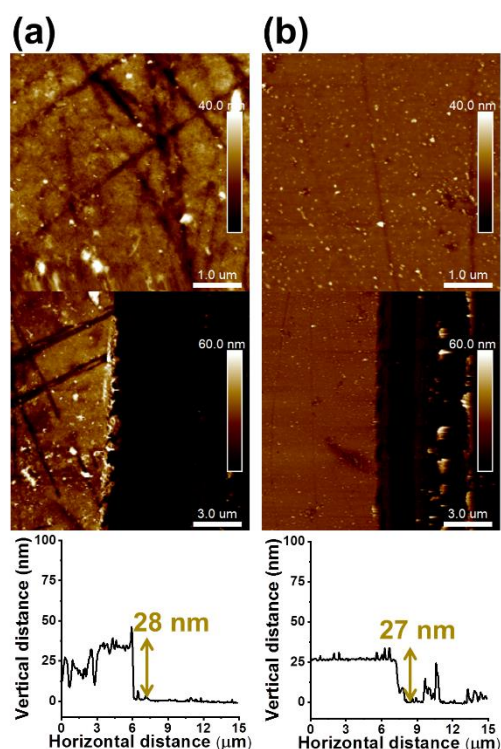


Figure S4. AFM analysis of polymer brushes grafted from a silicon surface *via* low ppm ATRP. Height images and cross-section profiles captured for homopolymer (a) PDMAEMA and (b) PtBA brushes. Synthesis conditions for: a) $[\text{DMAEMA}]:[\text{EBiB}]:[\text{Cu}^{\text{II}}\text{Br}_2/2\text{TPMA}] = 2400:1:0.72$; solvent used was $\text{EtOH}:\text{H}_2\text{O} = 2:3$; monomer volume = $50\%V_{\text{tot}}$; $V_{\text{tot}} = 6 \text{ ml}$; $t = 2 \text{ h}$; $T = \text{RT}$; blue light with $\lambda_{\text{max}} = 460 \text{ nm}$ was used as reducing factor; b) $[\text{tBA}]:[\text{EBiB}]:[\text{Cu}^{\text{II}}\text{Br}_2/2\text{TPMA}] = 1000:1:0.3$; solvent used was EtOH ; monomer volume = $50\%V_{\text{tot}}$; $V_{\text{tot}} = 6 \text{ ml}$; $t = 5 \text{ h}$; $T = 60^\circ\text{C}$; Cu wire was used as reducing factor ($S/V = 0.1$)

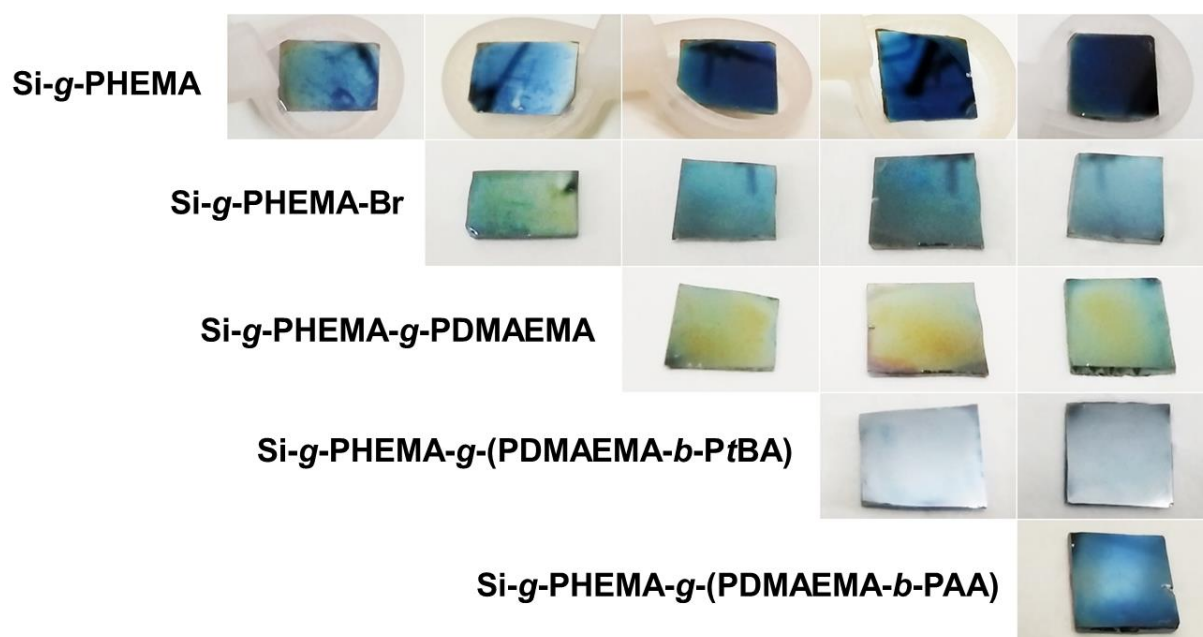


Figure S5. Summary of the real photographs of the Si wafers after the several stages of surface treatment (according to Scheme 1).

¹H NMR analysis

Additionally, the homopolymers (PDMAEMA and PtBA) were obtained in solution, purified through the dialysis using 1 kDa membrane, dried and characterized using ¹H-NMR. The chemical shifts identified in the spectrum presented in **Figure S5** (**Table 1**, entry 2), clearly evidenced PDMAEMA and protonated PDMAEMA [11] structures as follows: (500 MHz, D₂O) δ (ppm) = 0.78-1.10 (3H, -CH₃, α), 1.66-2.07 (2H, -CH₂-, β), 2.19-2.41 (6H, -N(CH₃)₂, c), 2.58-2.86 (2H, -CH₂-, a) 3.11-3.28 (6H, -N⁺(CH₃)₂, f), 3.54-3.69 (2H, -CH₂-, d), 3.92-4.26 (2H, -CH₂-, b), 4.33-4.52 (2H, -CH₂-, e).

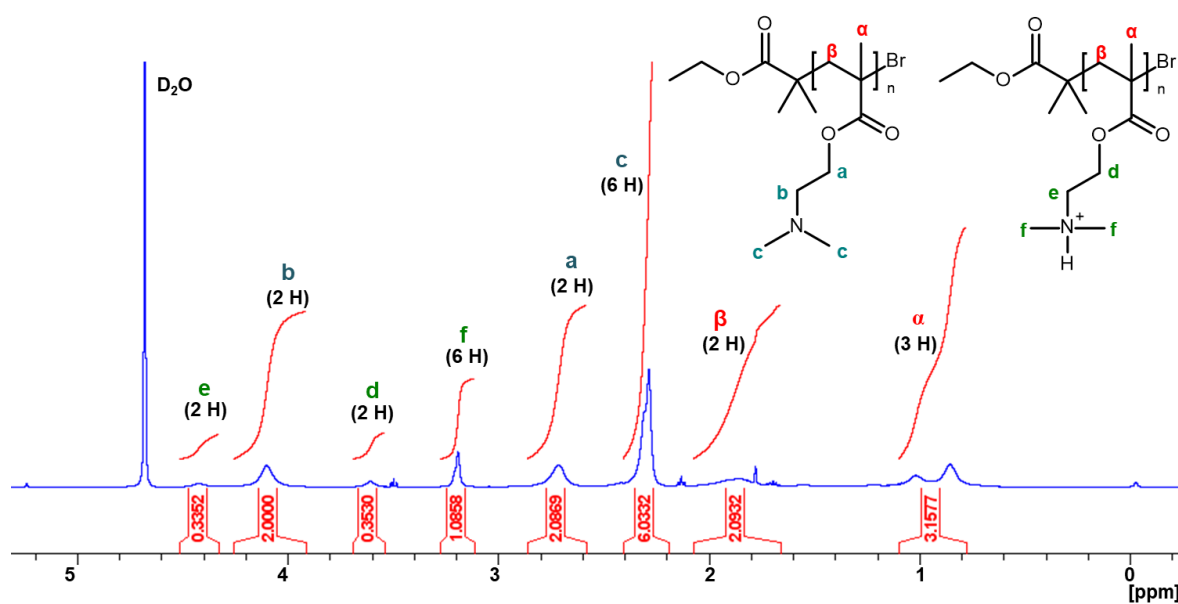


Figure S6. ^1H NMR spectrum of PDMAEMA (after purification) in D_2O (**Table 1**, entry 2).

The chemical shifts identified in the spectrum presented in **Figure S6** (**Table 1**, entry 3), clearly evidenced PtBA structure as follows: (500 MHz, CDCl_3) δ (ppm) = 1.35-1.47 (9H, $-\text{CH}_3$, a), 1.45-1.95 (2H, $-\text{CH}_2-$, β), 2.11-2.34 (1H, $-\text{CH}-$, α).

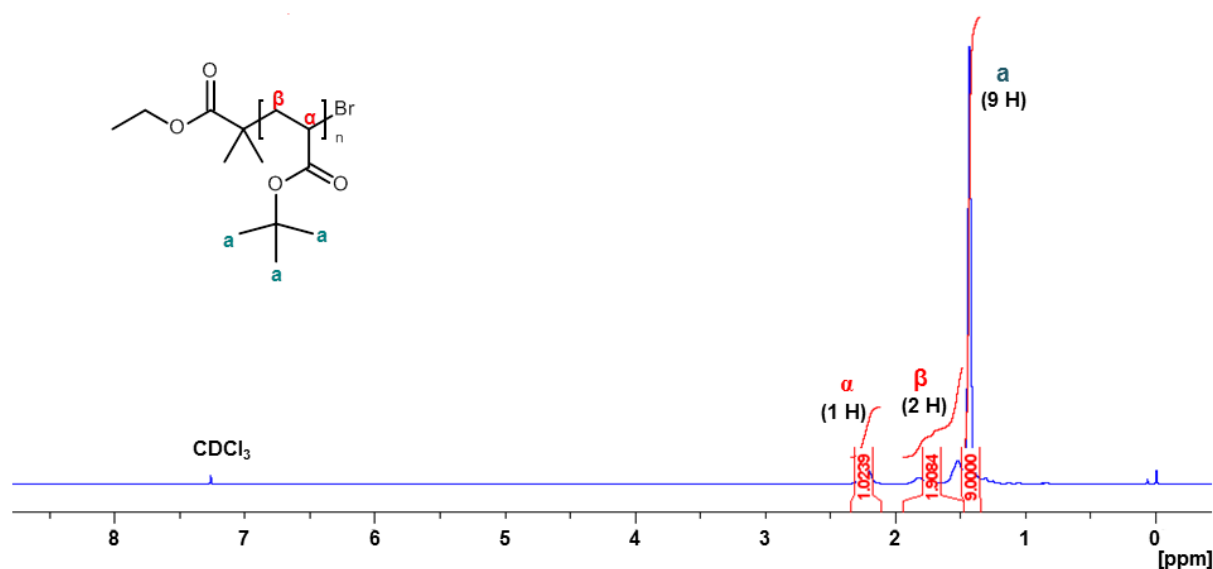


Figure S7. ^1H NMR spectrum of PtBA (after purification) in CDCl_3 (**Table 1**, entry 3).

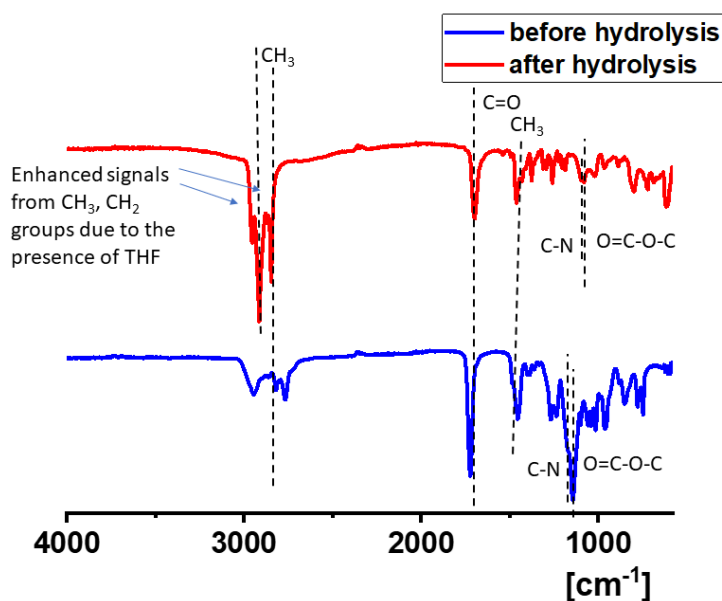


Figure S8. a) FT-IR spectra of PDMAEMA homopolymer before and after 24 h treatment in 10% TFA in DCM, measured in form of polymer coatings on silicon wafers.

S2.5 WCA, SEM, EDS, XPS and ToF-SIMS coating analysis

WCA and FSE analysis

Hydrophilic PHEMA-coated surface result in $\text{WCA} = 52^\circ \pm 2^\circ$, followed by a brominated polymer layer described by $\text{WCA} = 91^\circ \pm 1^\circ$, then the addition of a hydrophilic PDMAEMA segment caused $\text{WCA} = 39^\circ \pm 1^\circ$, but after addition of hydrophobic PtBA segment $\text{WCA} = 99^\circ \pm 2^\circ$. Finally, the hydrolysis procedure resulted in a hydrophilic nanolayer of complex architecture polymers characterized with $\text{WCA} = 31^\circ \pm 2^\circ$. The obtained WCA values were in agreement with the literature data determined for hydrophilic PHEMA [12], PDMAEMA [13] and PAA [14] as well as hydrophobic PtBA [15].

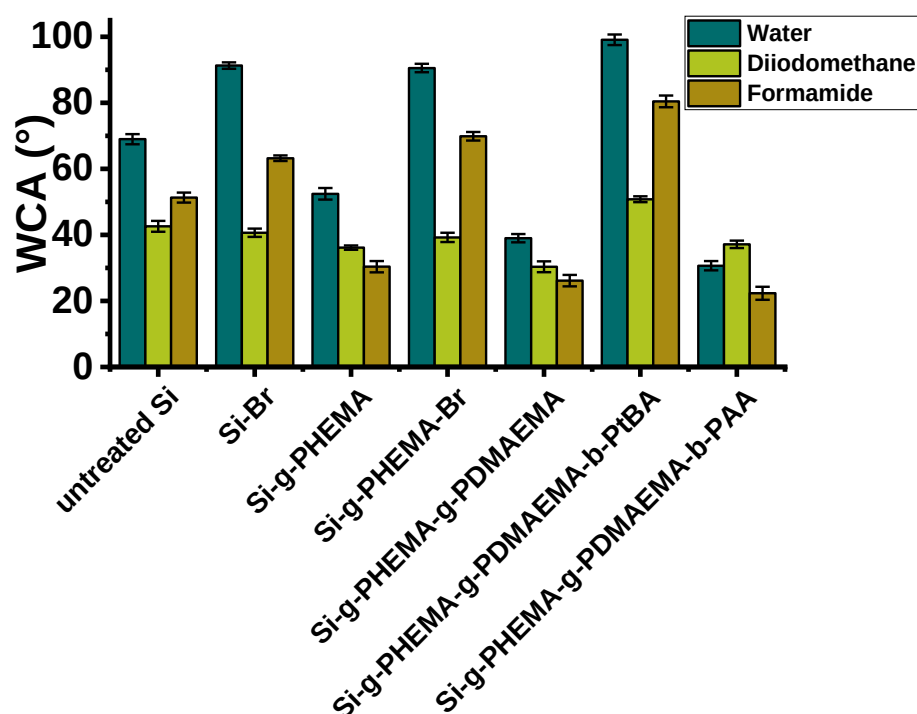


Figure S9. Experimental values of contact angles for each synthetic step according to Scheme 1, measured using polar (water, formamide) and non-polar (diiodomethane) liquids.

Table S4. Experimental values of contact angles for untreated Si wafers and modified with polymer brushes (according to Scheme 1).

Entry	Type of material	Experimental values of Θ [°]					
		Water	σ	Diiodomethane	σ	Formamide	σ
1	untreated Si	68.97	1.524	42.60	1.641	51.29	1.499
2	Si-Br	91.27	0.967	40.64	1.260	63.21	0.845
3	Si-g-PHEMA	52.42	1.747	36.14	0.640	30.37	1.698
4	Si-g-PHEMA-Br	90.51	1.269	39.23	1.414	69.85	1.302
5	Si-g-PHEMA-g-PDMAEMA	38.99	1.257	30.32	1.645	26.13	1.732
6	Si-g-PHEMA-g-(PDMAEMA-b-PtBA)	99.08	1.606	50.78	0.893	80.41	1.773
7	Si-g-PHEMA-g-(PDMAEMA-b-PAA)	30.65	1.406	37.12	1.129	22.32	1.971

Table S5. Parameters of free surface energy (FSE) calculated by the Owens-Wendt method for brominated surface and material modified with grafted brushes.

Entry	Type of material	Parameters of FSE [mJ/m ²]		
		water-diiodomethane		
		γ_s	γ_s^d	γ_s^p
1	untreated Si	42.19±0.1168	31.640±0.10633	10.553±0.0825
2	Si-Br	39.46±0.0954	38.6162±0.09539	0.8487±0.00269
3	Si- <i>g</i> -PHEMA	52.03±0.0936	31.3999±0.04390	20.6292±0.08268
4	Si- <i>g</i> -PHEMA-Br	40.15±0.1071	39.2403±0.10702	0.9142±0.00365
5	Si- <i>g</i> -PHEMA- <i>g</i> -PDMAEMA	60.59±0.1118	31.6285±0.07830	28.9638±0.07979
6	Si- <i>g</i> -PHEMA- <i>g</i> - (PDMAEMA- <i>b</i> -PtBA)	34.47±0.0769	34.2493±0.07690	0.2176±0.00102
7	Si- <i>g</i> -PHEMA- <i>g</i> - (PDMAEMA- <i>b</i> -PAA)	64.12±0.1049	27.4162±0.05588	36.7073±0.08881

SEM and EDS analysis

As a result of HEMA polymerization, a thin homogeneous layer can be observed in the Si-*g*-PHEMA SEM image (**Figure S10**), then after attachment of bromide molecules by ester groups, changes in surface morphology and the appearance of irregular pores. In the next step, a homogeneous highly packed PDMAEMA layer was obtained, testifying to the vertical position of the polymer molecules, and a similar image was registered after the addition of another polymer block in the form of PtBA. Finally, after hydrolysis of grafted polymer brushes, a slightly rougher surface with spherical formations indicating the interaction of carboxylic groups present in the PAA structure was observed on the Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA) surface [74].

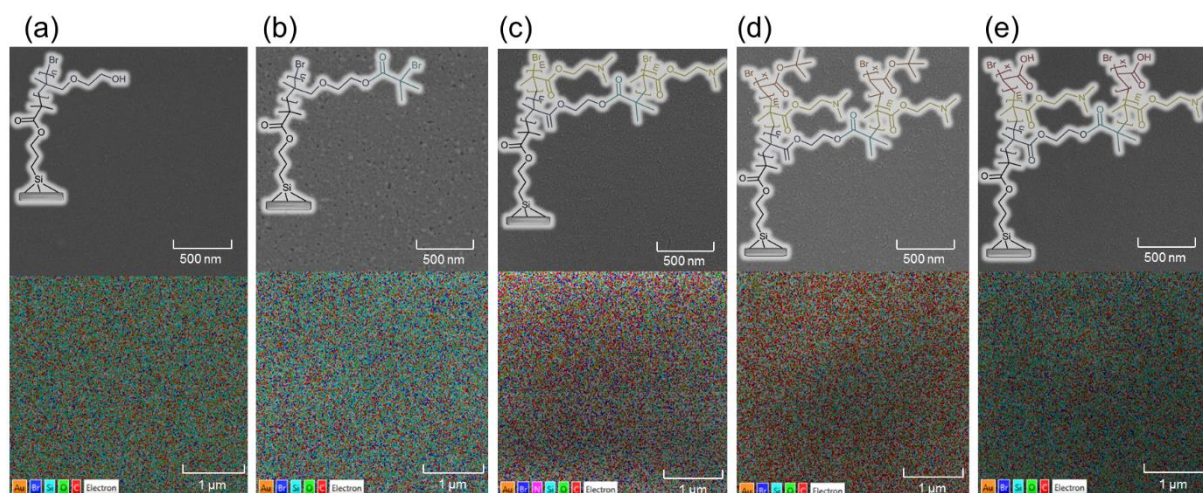


Figure S10. SEM image and elemental mapping of (a) Si-*g*-PHEMA; (b) Si-*g*-PHEMA-Br; (c) SEM image for Si-*g*-PHEMA-*g*-PDMAEMA; (d) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PtBA); (e) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA).

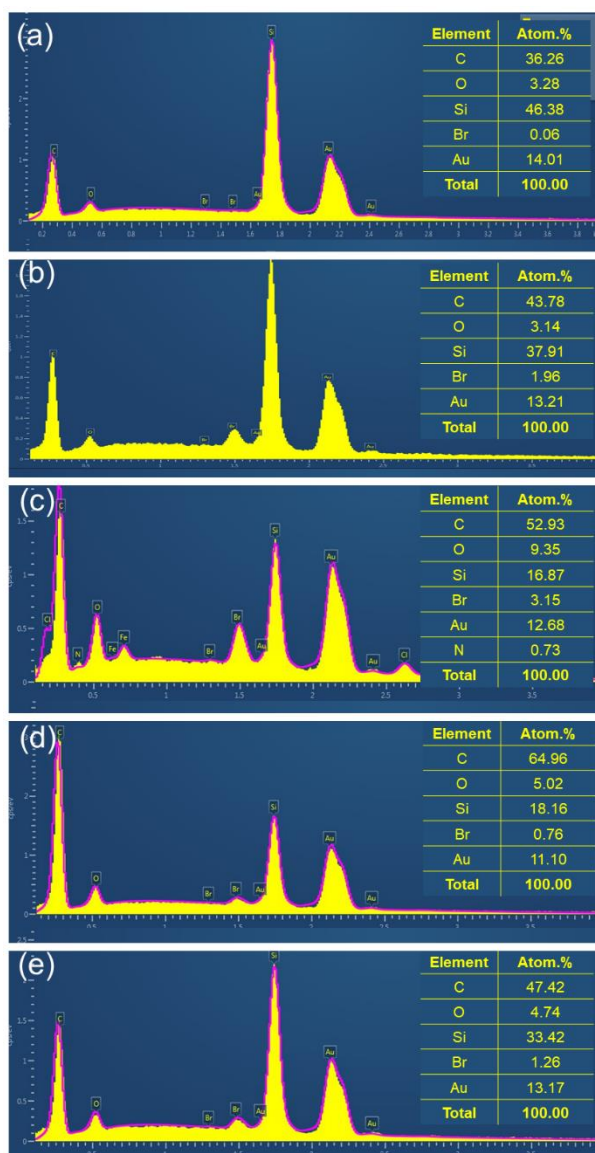


Figure S11. EDS spectrum for (a) Si-*g*-PHEMA; (b) Si-*g*-PHEMA-BiBB; (c) Si-*g*-PHEMA-*g*-PDMAEMA; (d) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PtBA); (e) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA).

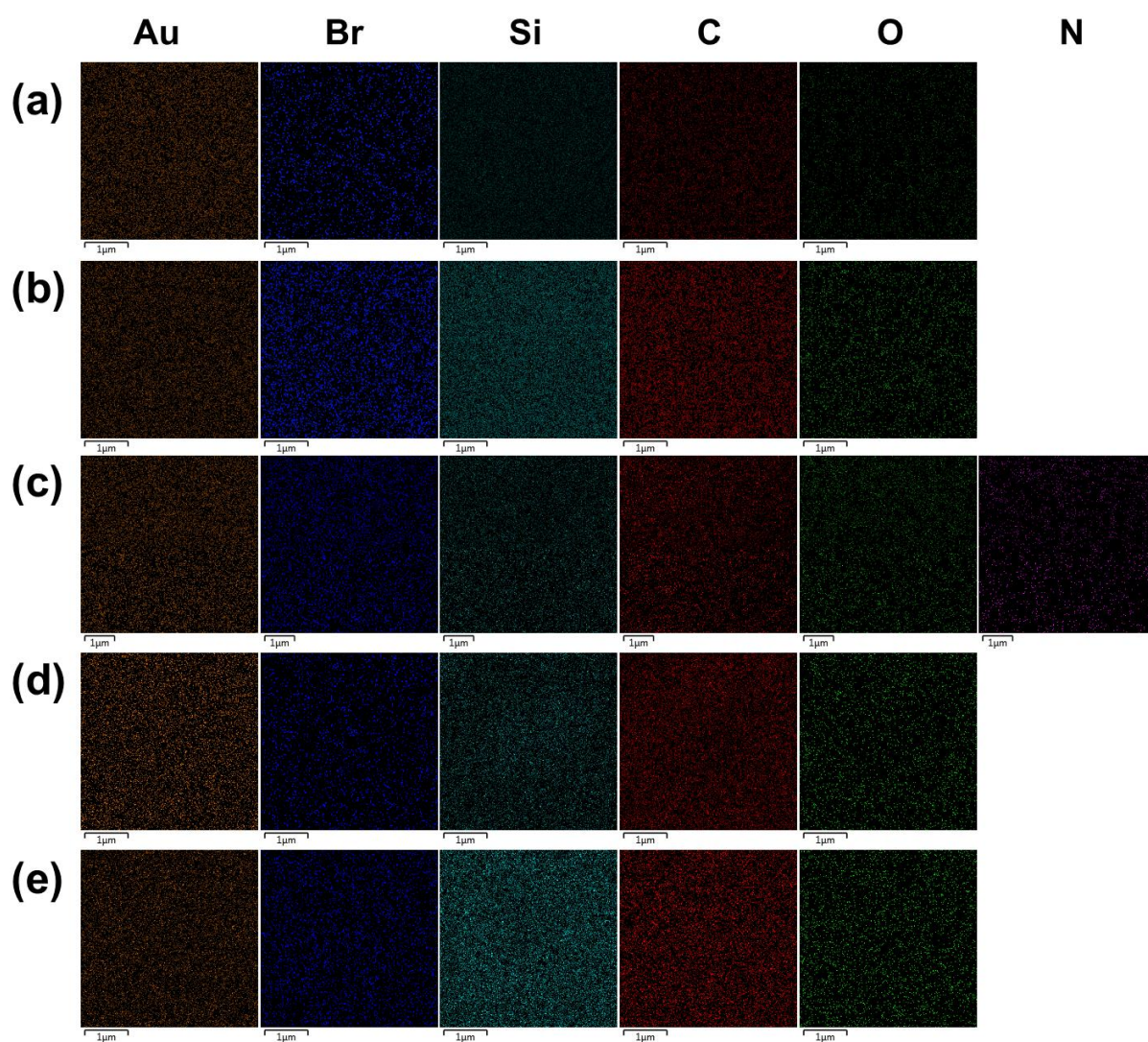


Figure S12. Elemental composition and their distribution of (a) Si-*g*-PHEMA; (b) Si-*g*-PHEMA-Br; (c) Si-*g*-PHEMA-*g*-PDMAEMA; (d) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PtBA); (e) Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA).

XPS analysis

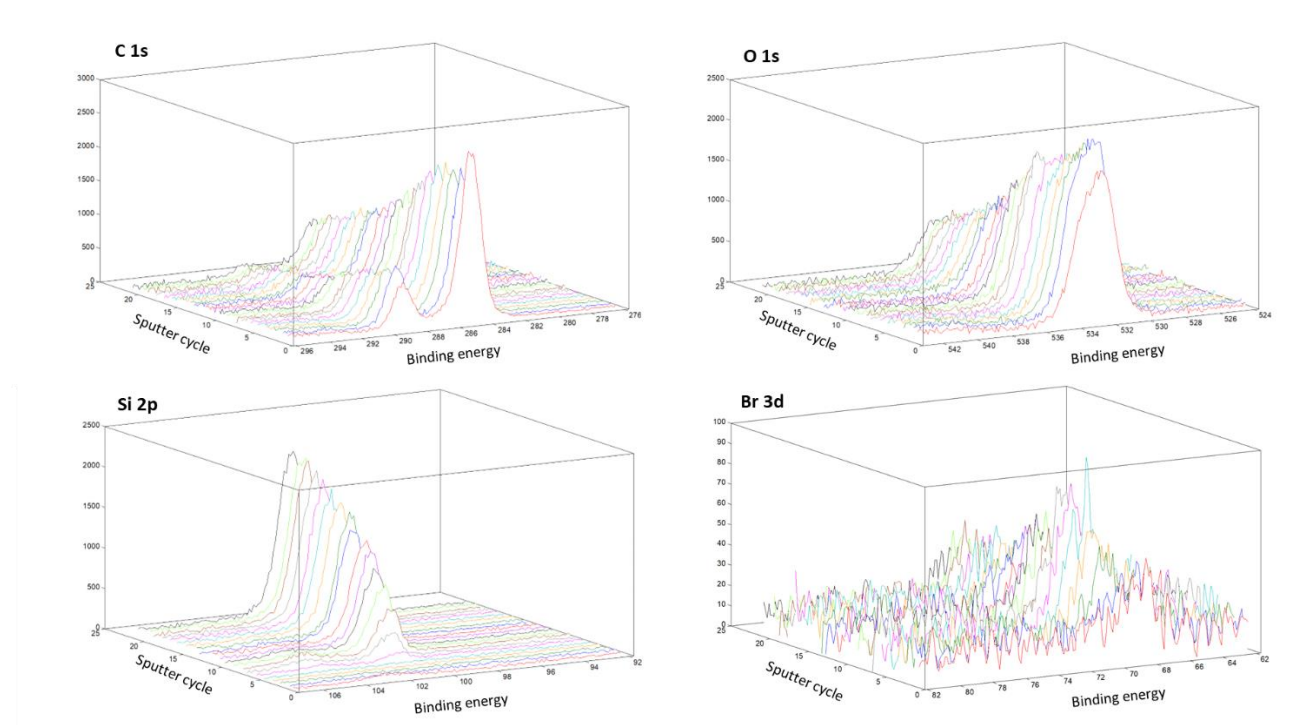


Figure S13. XPS depth-profiling data for Si-g-PHEMA-g-(PDMAEMA-*b*-PAA) synthesized according to **Scheme 1**.

Table S6. XPS elemental analysis of silicon wafer modified by the initiator (Si-Br). Silicon wafer grafted with PHEMA (Si-*g*-PHEMA generated according to **Table 1**, entry 1) and silicon wafer grafted with PHEMA brushes esterified via α -bromoisobutryl bromide (Si-*g*-PHEMA-Br) after presputtering. Percentage of total atomic concentration of detected elements analyzed from two independent points expressed as a mean $\pm$ std.dev.

Type of material	C	O	Si	Br
	% at.			
Si-Br	33.5 ± 2.8	33.5 ± 1.4	33.0 ± 1.4	0.0 ± 0.0
Si- <i>g</i> -PHEMA	68.8 ± 0.3	30.8 ± 0.2	0.4 ± 0.0	0.0 ± 0.0
Si- <i>g</i> -PHEMA-Br	71.0 ± 0.1	25.9 ± 0.2	0.6 ± 0.4	2.5 ± 0.4

Table S7. XPS elemental analysis of Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA) generated during depth-profiling with deconvoluted bond types (% at.).

Sputter time	C				N	O		Si	Br
	C-C	C-Br, C-C=O	C-O	COO		O-C	O=C, O-Si	Si oxide	
0	47.7	9.9	1.9	12.9	0.3	10.7	16.4	0.0	0.3
5	33.5	14.3	1.6	16.4	0.4	14.7	18.6	0.1	0.3
10	33.0	16.2	1.5	16.0	1.4	14.0	17.7	0.0	0.2
15	34.5	16.2	2.9	15.3	0.0	13.8	17.0	0.0	0.4
20	32.1	18.3	4.1	14.1	1.2	13.8	15.9	0.1	0.5
25	32.6	17.5	4.6	14.3	0.0	14.2	15.7	0.6	0.5
30	28.9	18.2	5.0	12.6	0.4	16.6	14.6	3.4	0.5
35	30.0	15.6	4.7	12.2	0.0	16.5	13.8	6.9	0.4
40	31.2	12.9	4.6	12.0	0.9	14.1	13.5	10.5	0.4
45	30.4	13.1	4.3	10.3	0.0	13.8	12.8	14.9	0.4
50	27.8	14.9	3.7	10.1	0.0	12.1	13.0	18.2	0.3
55	30.1	10.9	4.2	9.8	1.2	10.5	12.0	21.1	0.3
60	29.3	11.4	3.4	8.3	0.0	10.0	12.6	24.7	0.4
65	32.2	7.4	3.5	8.6	0.0	9.4	11.4	27.2	0.3
70	29.2	8.6	2.8	7.3	0.3	9.0	11.4	31.1	0.3
75	31.2	5.9	2.8	8.1	0.4	8.0	11.0	32.3	0.3
80	29.3	7.0	3.0	8.5	0.0	7.9	9.7	34.5	0.2
85	28.7	5.7	2.4	6.4	1.8	7.2	9.5	38.0	0.3
90	26.4	7.0	2.2	5.6	1.5	7.1	9.2	40.6	0.4
95	27.3	4.9	2.4	5.9	0.0	7.2	8.7	43.3	0.3
100	31.4	0.0	2.9	6.8	0.0	5.9	8.7	44.1	0.2

ToF-SIMS analysis

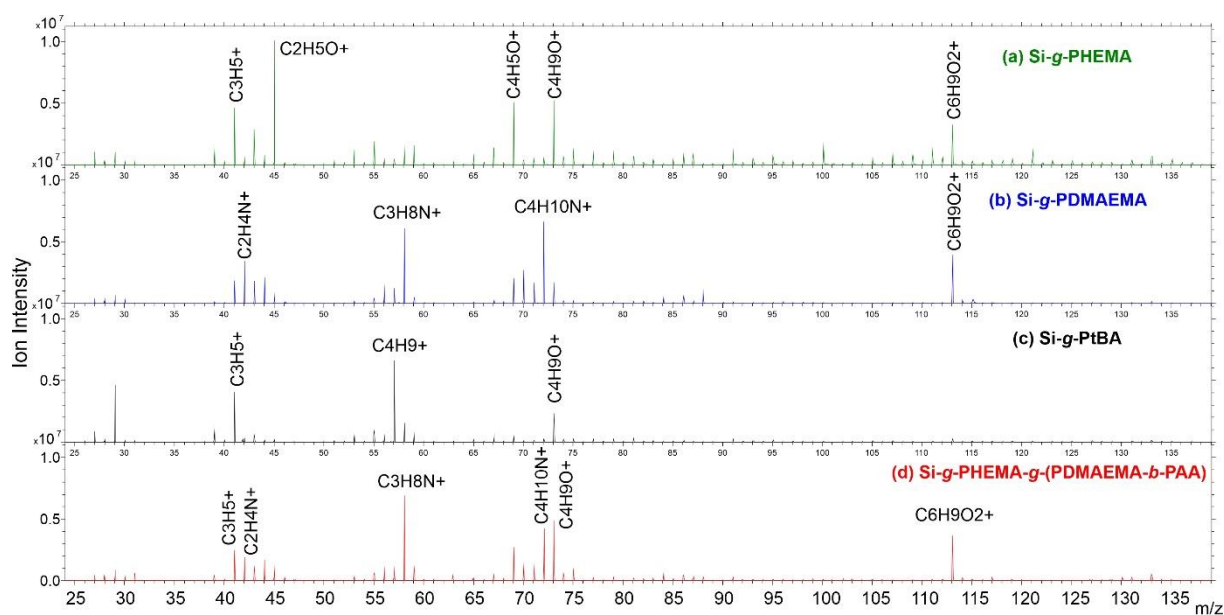


Figure S14. Representative positive ions ToF-SIMS spectra of Si-g-PHEMA (a), Si-g-PDMAEMA (b), Si-g-PtBA (c) and Si-g-PHEMA-g-(PDMAEMA-*b*-PAA) (d). Secondary ions of C_3H_5^+ , $\text{C}_2\text{H}_5\text{O}^+$, $\text{C}_4\text{H}_5\text{O}^+$, and $\text{C}_6\text{H}_9\text{O}_2^+$, characteristic for PHEMA (a); $\text{C}_2\text{H}_4\text{N}^+$, $\text{C}_3\text{H}_8\text{N}^+$, $\text{C}_4\text{H}_{10}\text{N}^+$, and $\text{C}_6\text{H}_9\text{O}_2^+$, typical for PDMAEMA (b) and C_3H_5^+ , C_4H_9^+ , and $\text{C}_6\text{H}_9\text{O}_2^+$, representative for PtBA (c) may be observed. In turn, for Si-g-PHEMA-g-(PDMAEMA-*b*-PAA), the series of peaks characteristic for PHEMA and PDMAEMA are visible whereas the C_4H_9^+ peak, typical for PtBA may no longer be observed confirming the successful hydrolysis of PtBA to PAA.

S.2.6 WCA and AFM polyelectrolytes characteristic

Table S8. Results of water contact angles measurements for silicon wafers grafted with homopolymers (Si-*g*-PDMAEMA, Si-*g*-PAA) and complex architecture polymers (Si-*g*-PHEMA-*g*-(PDMAEMA-*b*-PAA)), using solutions characterized by various pH in decreasing and increasing manner.

Type of material	Si- <i>g</i> -PDMAEMA				Si- <i>g</i> -PHEMA- <i>g</i> -(PDMAEMA- <i>b</i> -PAA)				Si- <i>g</i> -PAA			
pH	WCA Θ decr [°]	σ decr [°]	WCA Θ incr [°]	σ incr [°]	WCA Θ decr [°]	σ decr [°]	WCA Θ incr [°]	σ incr [°]	WCA Θ decr [°]	σ decr [°]	WCA Θ incr [°]	σ incr [°]
2	60.36	0.97	59.19	1.93	70.63	1.83	70.56	1.41	56.77	1.78	57.84	1.67
4	71.15	1.26	72.85	1.14	53.24	1.79	67.93	1.62	50.18	1.80	56.01	1.89
5	74.06	1.30	78.36	1.34	47.92	1.21	66.37	1.86	48.6	1.56	54.6	1.39
6	79.50	1.68	80.46	0.91	33.95	1.79	64.63	1.78	48.19	1.51	53.1	1.68
7	79.41	1.58	80.23	1.22	23.57	1.84	61.07	0.78	47.48	1.36	51.0	1.64
10	73.66	1.65	76.85	1.42	17.40	1.54	47.38	1.54	45.97	1.27	46.62	1.64
12	70.45	1.69	70.95	1.54	17.31	1.86	17.46	0.87	38.91	1.09	38.38	1.79

S.2.7 μ L-scale step-by-step synthesis of complex architecture polymers

Rescaling all the synthesis steps to microliter scale allowed a step-by-step synthesis of a hydrophilic polyelectrolytes coating, characterized by the wetting angle equal to 49.9°. The differences in the WCA values for the individual steps comparing the synthesis scales are most likely due to differences in the thickness of the individual polymer blocks. The most notable is the difference in WCA after grafting PtBA (99.1° in mililiter scale whereas 78.4° in microliter scale), which might stem from lower resistance to oxygen as well as lower activity of acrylates in comparison with methacrylates. The obtained PtBA brushes were therefore presumably much shorter and resulted in lower value of WCA. Similarly after hydrolysis the PAA block was smaller which is the reason of less evident hydrophilic properties of final hybrid material.

Table S9. Comparison of WCA values for hybrid materials synthesized in mililiter and microliter scale.

Type of material	WCA Θ [°]	
	Mililiter scale	Microliter scale
Si- <i>g</i> -PHEMA	52.42±1.75	53.64±1.93
Si- <i>g</i> -PHEMA-Br	90.51± 1.27	80.97± 1.97
Si- <i>g</i> -PHEMA- <i>g</i> -PDMAEMA	38.99± 1.26	41.22± 1.92
Si- <i>g</i> -PHEMA- <i>g</i> -(PDMAEMA- <i>b</i> -PtBA)	99.08± 1.61	78.41± 1.78
Si- <i>g</i> -PHEMA- <i>g</i> -(PDMAEMA- <i>b</i> -PAA)	30.65± 1.41	49.91± 1.80

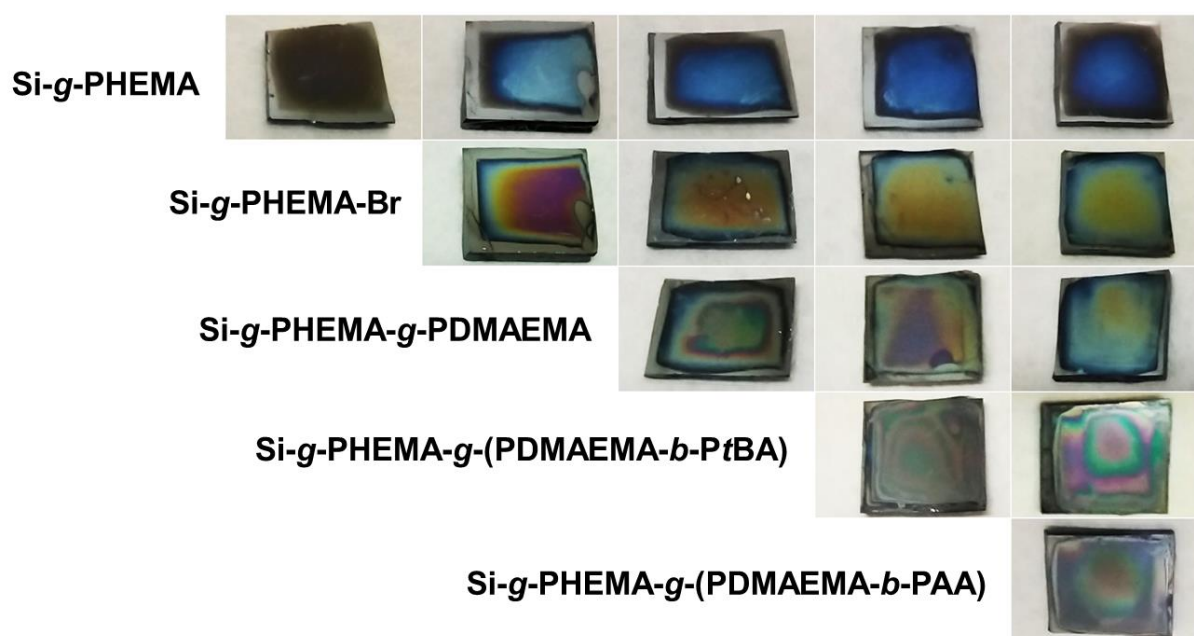


Figure S15. Summary of the real photographs of the Si wafers after the particular stages of μL -scale surface treatment.

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